DualCard User Manual

DUALi Inc.

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Our product development center is located in South Korea to provide technical support. For assistance, contact our technical support team below.

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Revision History

- 2009.02. (Ver. 1.0) : First Release
- 2009.05. (Ver. 1.01) : 15693 (Optional. DE-620R) Function Added
- 2010.04. (Ver. 1.02) : Mifare plus function tap added
- 2013.08. (Ver. 1.1) : Layout composition modified (DualCard version 3.1)
  - Update Script input more user-friendly
  - Mifare tab: 4/7/10 bytes UID support function added
  - NFC tab : LLCP/SNEP (ISO 18092) and NFC Barcode test function added.
  - UTIL tab : device number check, serial communication speed select function, Triple DES function added.
- 2015.03.11 (Ver.2.0) : DualCard layout composition modified. (DualCard version 4.0)
  DESFire function improved. (Batch function)
  Util tab revision (Util 1/ Util 2)
  PCSC mode test function (all tabs)
- 2016.04.12 (Ver.2.1) : MIFARE Ultralight C tap added (Ch.4.4)
  Contents modified
  Felica, DesFire features improved.
- 2018.07.31 (Ver.2.2): Layout Update

Table of Contents

Introduction

A. Intended Readership

This manual was written while keeping in mind the end-user. This manual is intended to act as a reference.

B. Purpose

This manual describes the various functions that will be provided in the application. The purpose of this document is to give detailed descriptions about the Application Controls and the Interfaces with readers and cards.

C. Scope

The DualCard program supports the following DUALi readers, which support the following tags and cards.

Tags and Cards

- ISO 7816
- ISO 14443 Part 3 and Part 4 (Type A and Type B)
- ISO 15693
- NFC
- Mifare Series (registered trademarks of NXP Semiconductors)
- DesFire (registered trademarks of NXP Semiconductors)
- Felica (registered trademark of SONY Corporation)

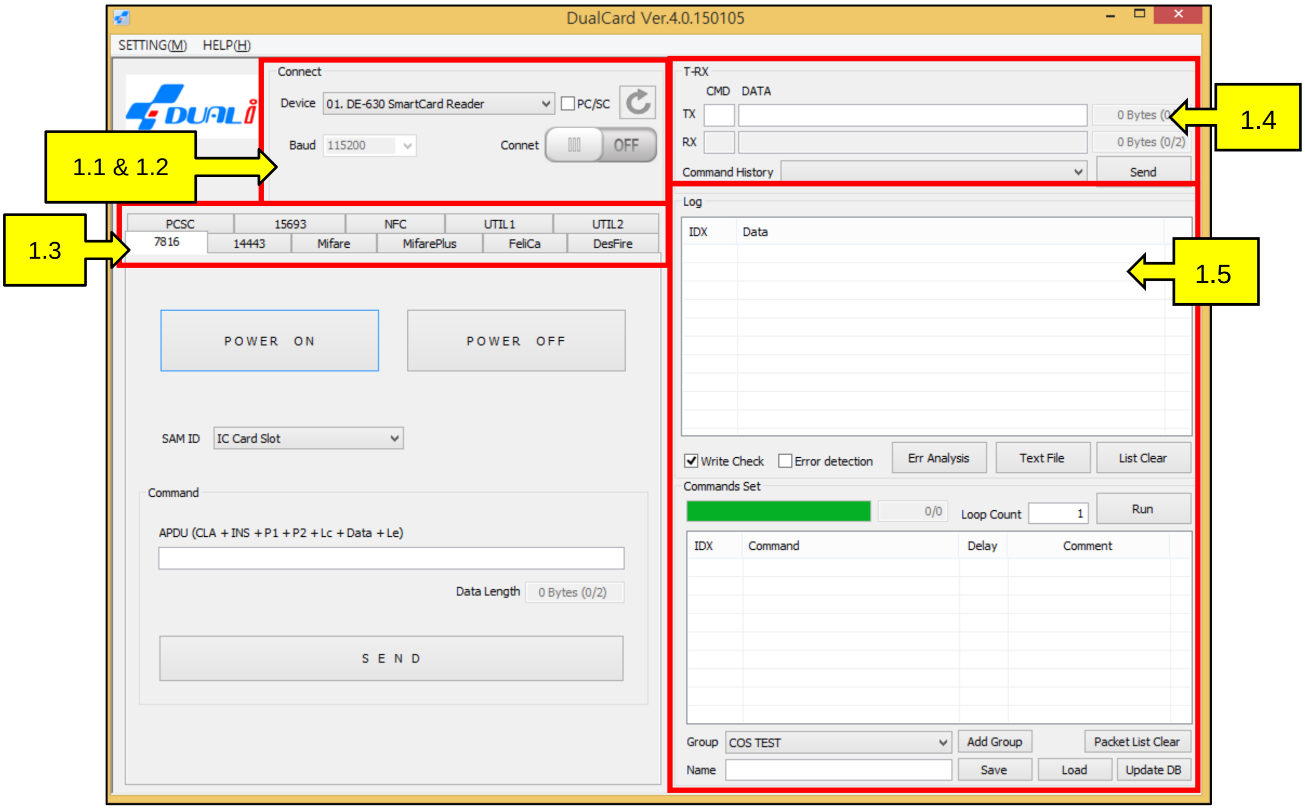
DUALi Readers

- DE-620/620R/620V/620A, DE-630/631, Dragon, DE-930/950 FDK, NFC PAD
- All DUALi modules

Supported Operating Systems:

Windows 2000, Windows XP, Windows Vista (32bit), Windows 7 (32/64bit), Windows 8, Windows 8.1, Windows 10

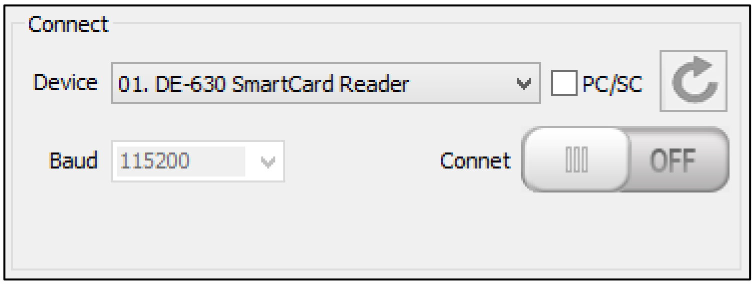
1. DualCard Layout



< Figure 1-1 >

DualCard Main Layout

1.1 Connect (Vender Mode)



<Figure 1-5>

Connect, Vendor Mode

Once a device is connected, the product name is displayed on the “Device” dropdown box. The support interface can either be USB or Hyper Terminal (RS-232). If UART is used, the standard communication speed will be set to 115200bps.

– 

: Click to start a test of the reader, once a reader is selected from the "Device" list.

– : Click to refresh the list of devices when an additional reader is connected while a test is running.

– 

: Status of a connection changes to "ON" when a device is connected

DualCardDll API: DE_InitPort([PORT],[BAUD]);

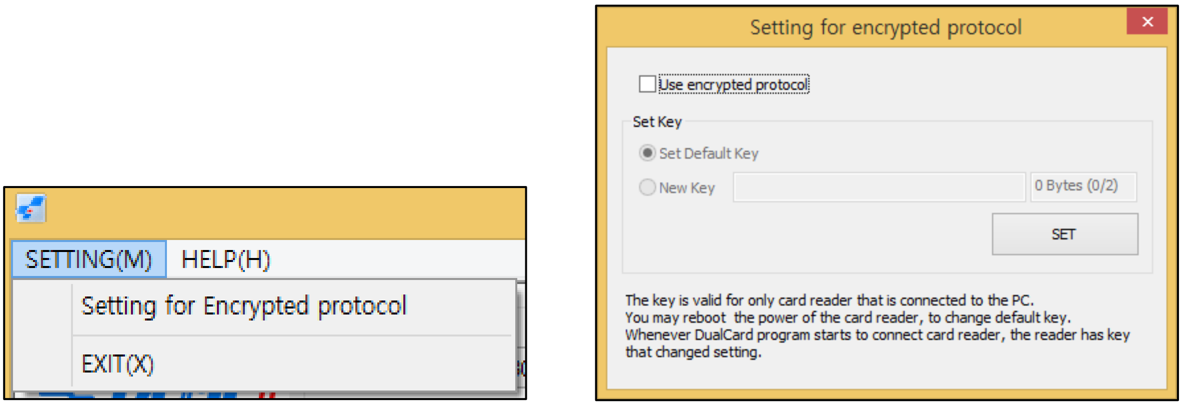
– 

: Status of a connection changes to "OFF" when a device is disconnected

DualCardDll API: DE_ClosePort(m_pDoc->m_nPort);

You are able to use encrypted protocols between a PC and a DUALi device with the DualCard Version 4.0. To use this function, select "Setting for Encrypted protocol" from the main menu._

*To use this function, you must update the firmware. Contact the DUALi Lab for assistance.

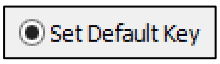


<Figure 1-3>

Encrypted Protocol

– 

: Check the box to activate encrypted communication

– 

: Select this option to use the default key in the terminal for encrypted communications. The reader is set to the default key after it has been reset.

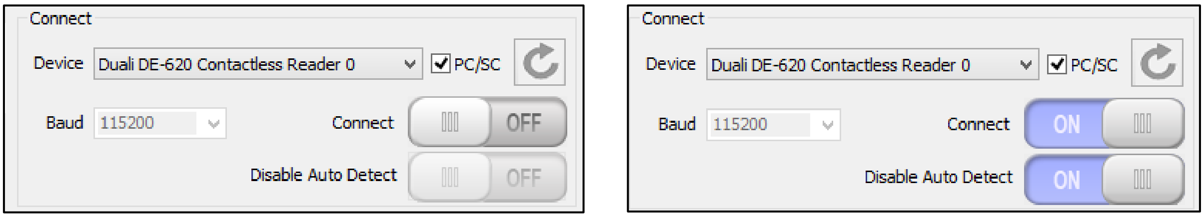
- 

: Select this option to use the User's key for encrypted communication. The new key setting will be valid after exiting and reopening the DualCard system. However, after the reader has been reset, the key setting will return to the default key.
- 

: Click to save settings. While this function is activated, you can find the 

under the DUALi logo in program in the main menu, <Figure 1-1>.

1.2 Connect (PC/SC mode)



<Figure 1-4>

Connect PC/SC Mode

- 

: Check the PC/SC box to have PC/SC devices populated in the device list
- 

: Click once you have placed the RF card on the reader
- 

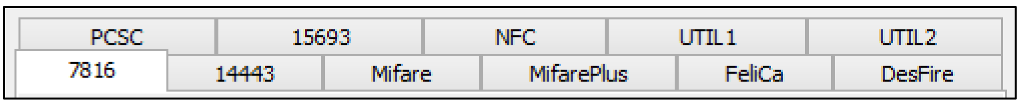
: Click to "OFF" to disable the auto detect function. This allows the use of all function tabs in DualCard.

DualCardDll API :

DE_SCardConnect([PORT]);

DE_SCardDisConnect(m_pDoc->m_nPort);

1.3 Function Tabs

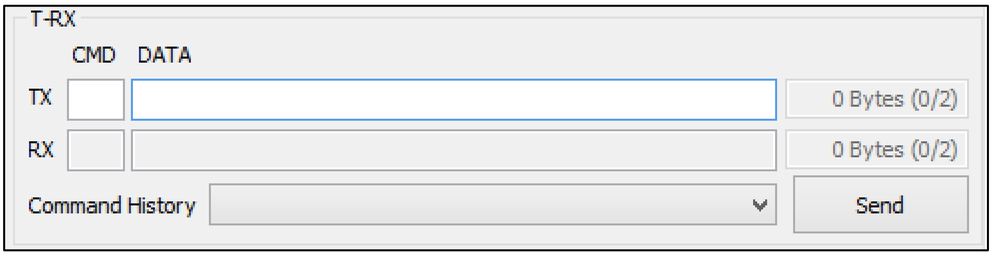


<Figure 1-5>

Function Tabs

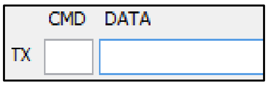
- The figure above shows the various protocols offered in the system.
- DualCard supports ISO 7816, ISO14443 A/B, Mifare, MifarePlus, FeliCa, DesFire, PCSC, ISO15693, NFC and Utility funtions. Please double check the protocols that the device supports.

1.4 Command Input



<Figure 1-6>

T-RX

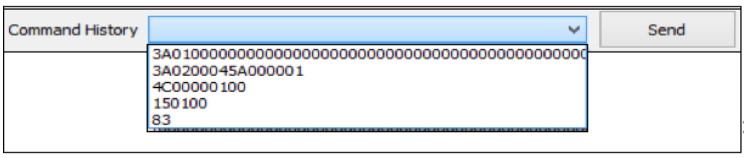
- The T-RX group box, <Figure 1-6>, is for the user to input direct commands from the protocol specifications. Each command must be no more than 1 byte size (as Hexadecimal).
- 

: Input the command (2 digit Hexadecimal) in the first field and the data in the second field. Then click "Send".
- 

: Click to execute a command

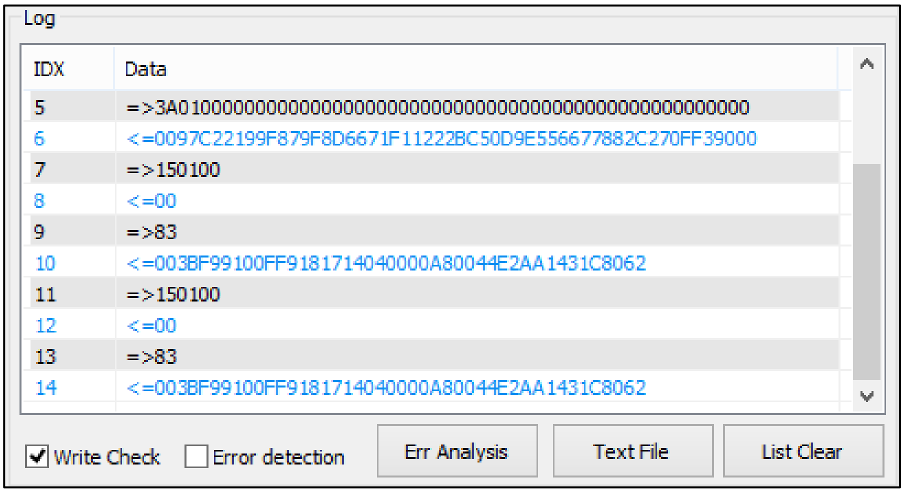
  DualCardDll API: DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF] ,[TOUT])

- 

: The response value from reader/card is displayed here. The response code is displayed in the first field and the response data in the second field.
- 

: The Dropbox displays the 10 most recent commands in order.

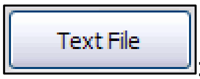
1.5 Packet List



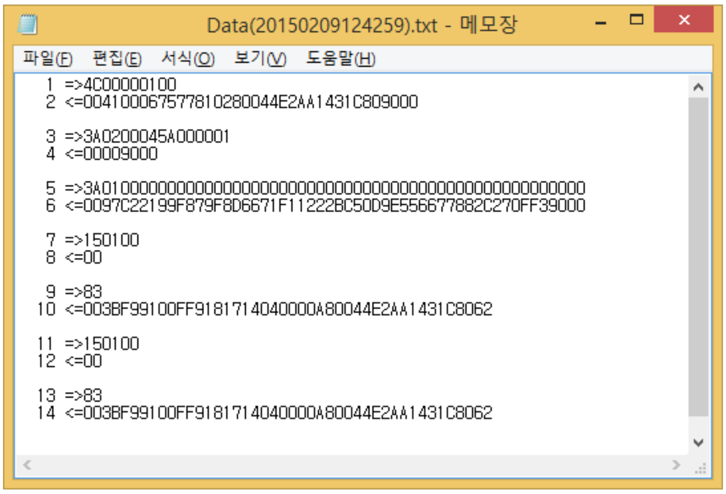
<Figure 1-7>

Log

This section displays the Send and Received Data List. It shows all of the communication data between the PC and device. By double-clicking a command, from the command set <Figure 1-10>, you can add the selected command to the packet list.

– 

: Click to make the list of commands and responses as a txt. file, as shown in <Figure 1-8>. The file name is automatically generated with the date and time. See example below, **Data(20081222140537).txt**. The file is saved in the)DualCard\textard\text

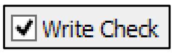


<Figure 1-8>

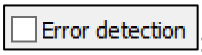
Text File

– 

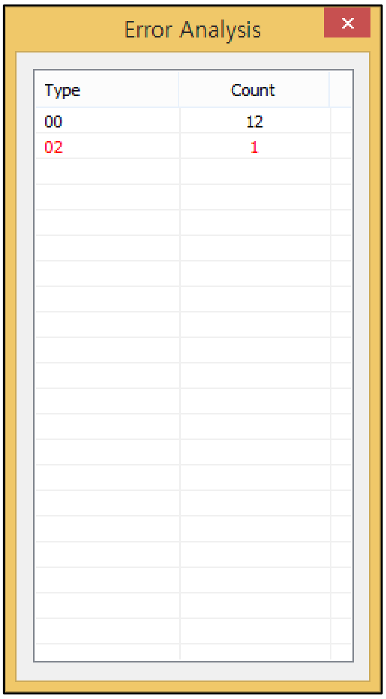
: Click to delete all of the commands and responses in the log.

– 

: Check the box to display all communication data (Commands & Responses) in the Packet List, <Figure 1-10>. If it is not checked, the command and response in <Figure 1-7> will be not shown on the Packet List, <Figure 1-10>.

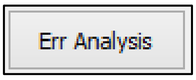
– 

: Check the box to stop loop testing when the response code detects an Error code.



<Figure 1-9>

Error Analysis

– 

: Click to see error information.

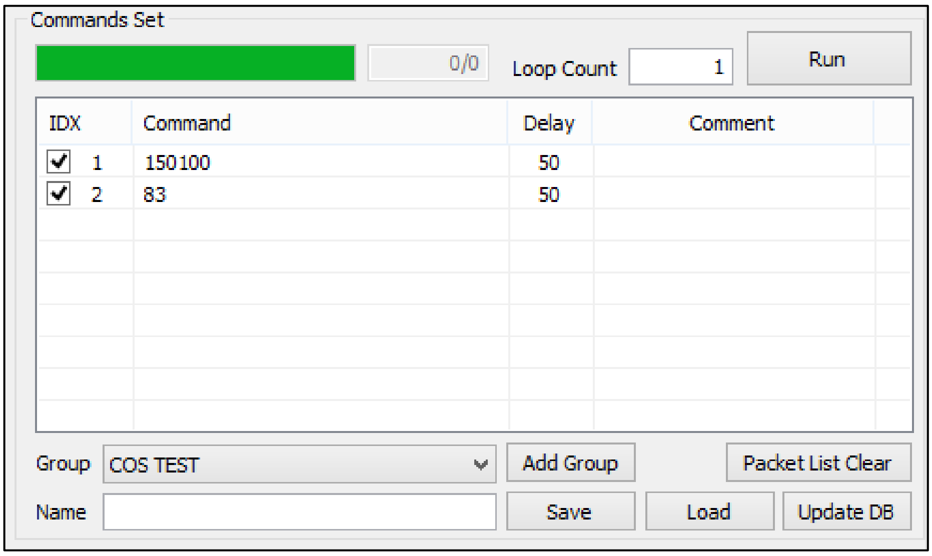
Error Information Codes:

00 : No Error

02 : No response from the card

03 : CRC Error

17 : Unknown command



<Figure 1-10>

Packet List

- Double click the commands from the Log, <Figure 1-7>. The selected commands will be added to the packet list, <Figure 1-10>.
- 

: Click to execute commands, the commands with the checked boxes will be executed the number of times that is in the “Loop Count” field, 

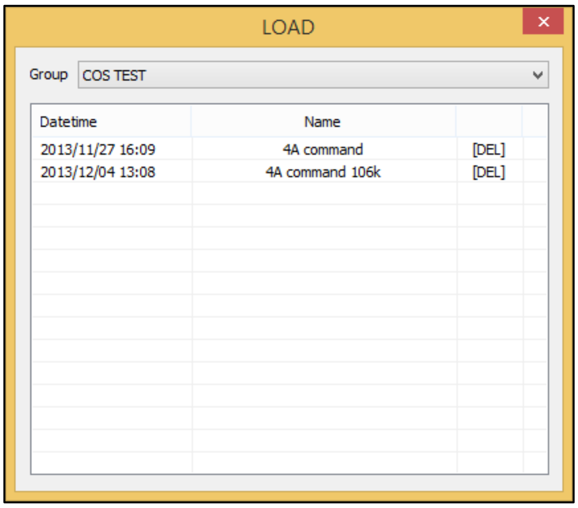
.
- 

: Click to clear the Packet List.
- 

: input a file name and click 

to save the full command list as a file. It will be saved to the packet list in the database.
- 

: Click to display a saved file to the list. The selected file will be displayed like in <Figure 1-11>.



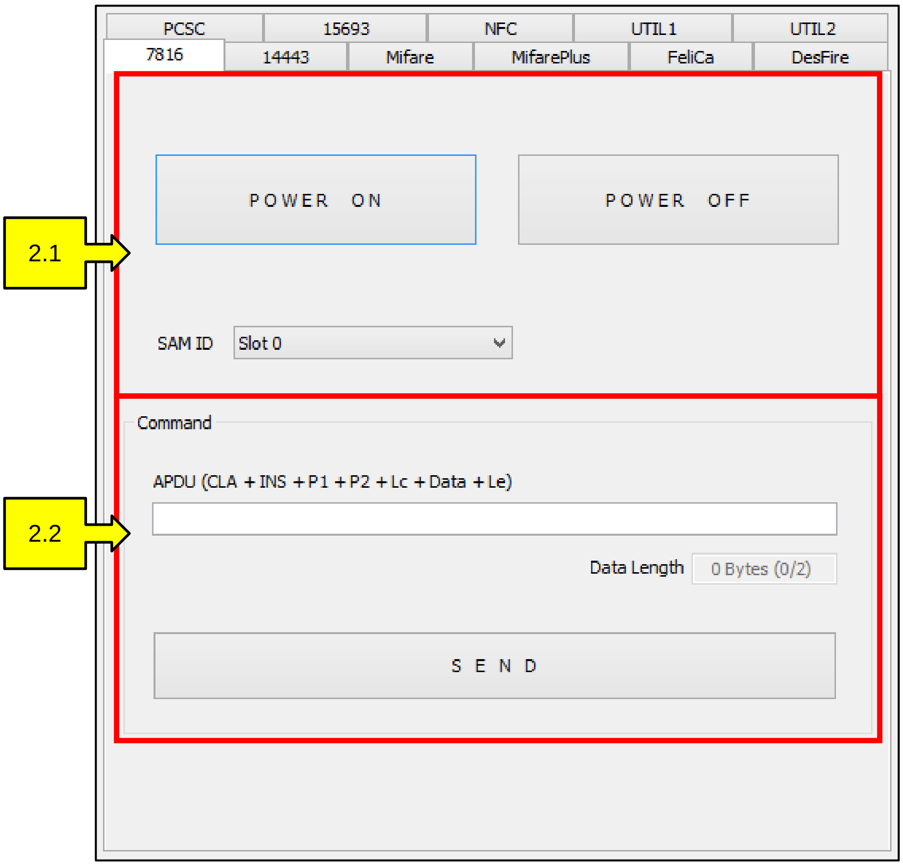
<Figure 1-11>

Load

- From <Figure 1-11>, double click on a command name in the “Name” field to add command sets to the packet list, <Figure 1-10>.
- From <Figure 1-11>, double click on “[DEL]”, next to the file name to delete a command set

2. 7816

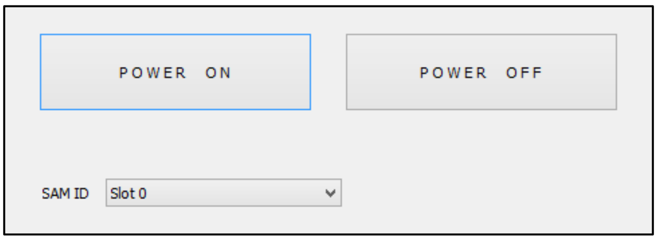
The 7816 tab, <Figure 2-1>, supports communication with ISO 7816 IC contact cards. From this tab, you can manage the Power On/Off function and command execution.



<Figure 2-1>

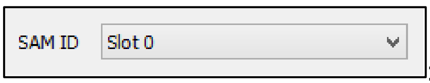
7816 Tab

2.1 Power On/Off



<Figure 2-2>

Power On/Off

– 

: Select the appropriate card slot to turn the power on/off to.

– POWER ON : Click to turn the power on for the card slot (SAM Slot/ IC Card Slot).

DualCardDll API : DE_IC_PowerOn([PORT], [SLOT NO], [RLEN], [RBUF]);

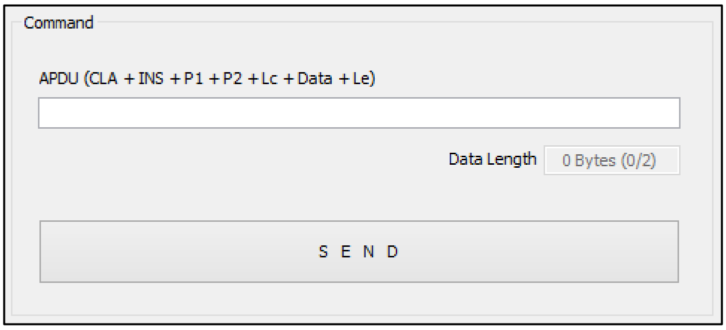
DUALi RW Protocol : C0[SLOT NO]00

– POWER OFF : Click to turn the power off for the card slot (SAM Slot/ IC Card Slot).

DualCardDll API : DE_IC_PowerOff([PORT], [SLOT NO]);

DUALi RW Protocol : C5[SLOT NO]

2.2 Command



<Figure 2-3>

7816 Command

– APDU (CLA + INS + P1 + P2 + Lc + Data + Le) : Input a command (digit Hexadecimal), that complies with ISO 7816, and click “Send”. A response will be shown on the log, <Figure 1-7>.

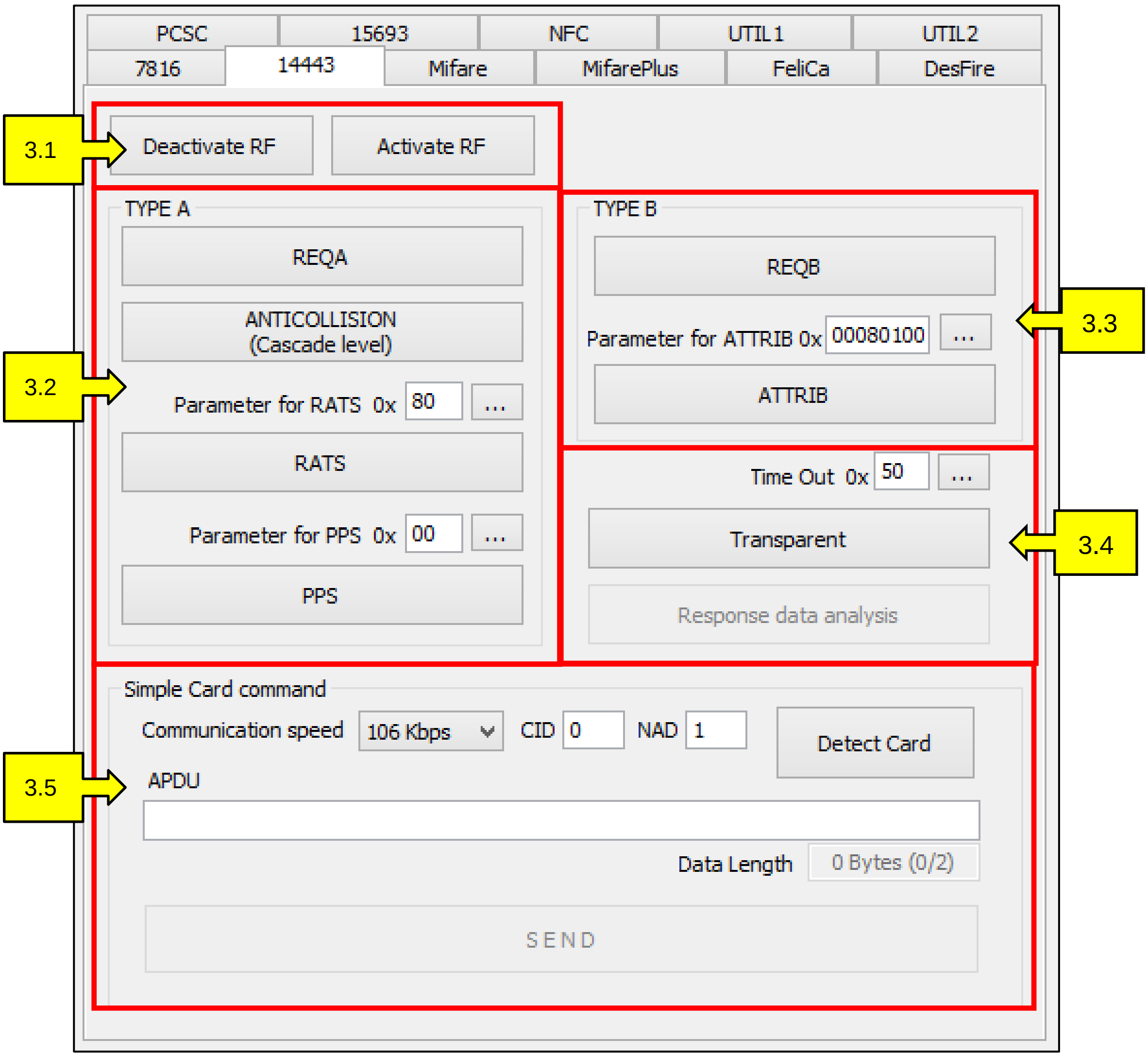
DualCardDll API : DE_IC_Case4 ([PORT], [SLOTNUM], [SLEN] , [SBUF], [RLEN], [RBUF]);

DUALi RW Protocol : C4[SLOT NO][APDU]

– S E N D : Click to execute a command.

3. 14443

The 14443 tab, <Figure 3-1>, supports communication with ISO 14443 A/B compliant cards. In this tab, you can manage the Type A command set, Type B command set, card detect function, User command transmit, and RF field control. For the technical detail of the protocol, refer to ISO 14443 Specifications.



<Figure 3-1>

14443 Tab

3.1 RF field Activation

– 

: Select to de-activate the RF field.

DualCardDll API : DE_RFOff([PORT]);

DUALi RW Protocol : 11

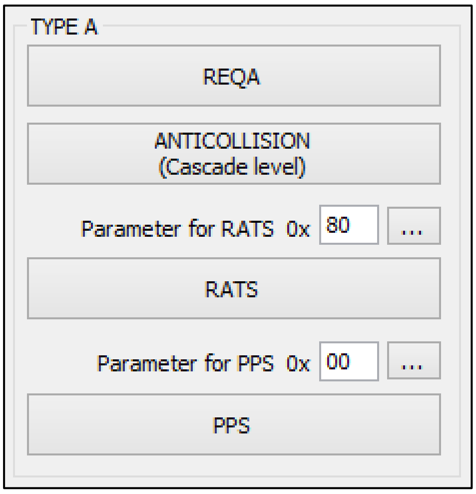
- 

: select to activate the RF field.

DualCardDll API :DE_RFOn([PORT]);

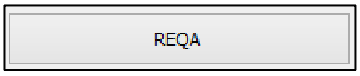
DUALi RW Protocol : 10

3.2 Type A



<Figure 3-2>

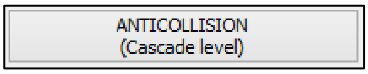
Type A

- 

: Click to execute the REQA command. A response will be shown on the log, <Figure 1-7>.

DualCardDll API : DEA_Idle_Req([PORT],[RLEN],[RBUF]);

DUALi RW Protocol : 21

- 

: Click to execute and get respon”, which supports cascade level commands. A response will be shown on the log, <Figure 1-7>.

DualCardDll API : DEA_AntiSelLevel([PORT],[RLEN],[RBUF]);

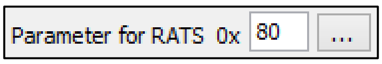
DUALi RW Protocol : 3D

- 

: Click to execute RATS commands. A response will be shown on the log, <Figure 1-7>.

DualCardDll API : DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF]);

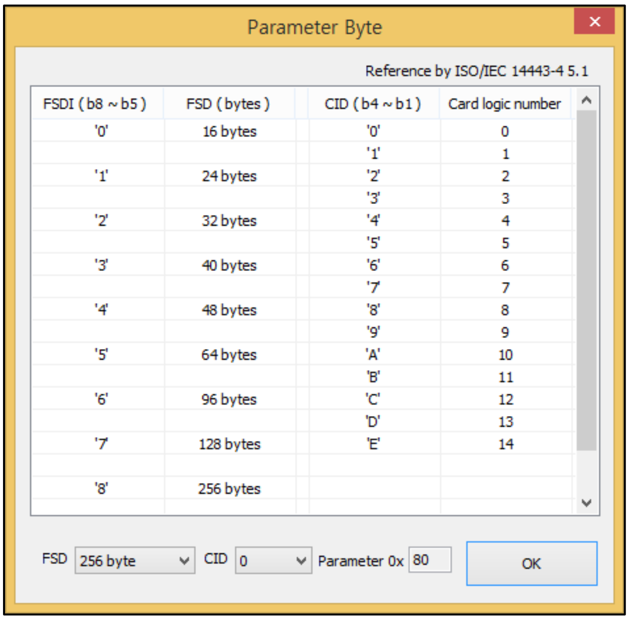
DUALi RW Protocol : 41E0[RATS][TIMEOUT]

- 

: Input the parameter byte for RATS or click the “…” button to automatically generate the value.

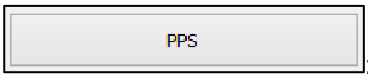
– 

: Click to display the parameter byte setting window for RATS, <Figure3-3>.



<Figure 3-3>

Parameter Byte, RATS

– 

: Click to execute the PPS Command. A response will be shown on the log, <Figure 1-7>.

DualCardDll API : DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF]);

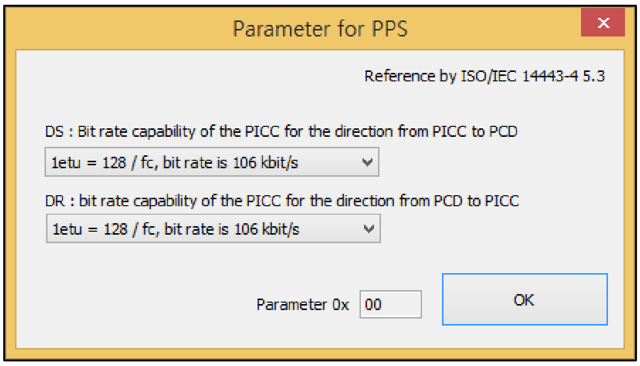
DUALi RW Protocol : 41[PPS]1100[TIMEOUT]

– 

: Input the parameter byte for PPS or click the "…" button to automatically generate the value.

– 

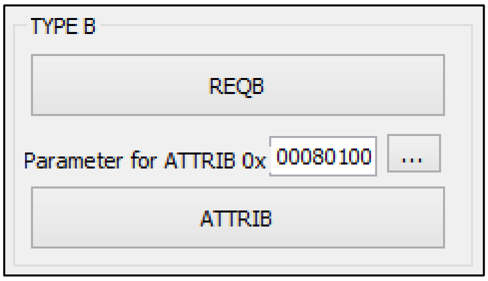
: Click to display the parameter byte setting window for PPS <Figure 3-4>.



<Figure 3-4>

Parameter Byte, PPS

3.3 Type B



<Figure 3-5>

Type B

– 

: Click to execute the REQB command. A response will be displayed in the log <Figure 1-7>.

DualCardDll API : DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF] ,[TOUT]);

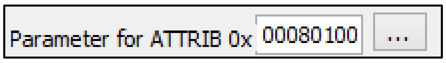
DUALi RW Protocol : 6005000050

– 

: Click to execute the ATTRIB command. A response will be displayed in the log <Figure 1-7>.

DualCardDll API : DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF] ,[TOUT]);

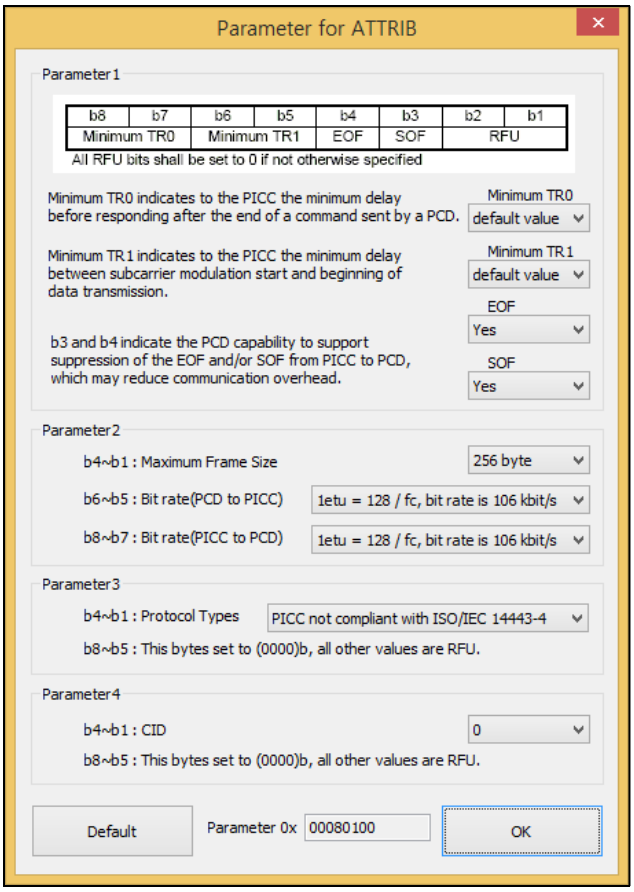
DUALi RW Protocol : 601D[ID][PARAMETER][TIMEOUT]

– 

: Input the parameter byte for ATTRIB or click the "…" button to automatically generate the value.

– 

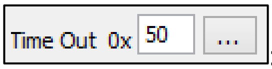
: Click to display the parameter byte setting window for ATTRIB, <Figure 3-6>.



<Figure 3-6>

Parameter Byte, ATTRIB

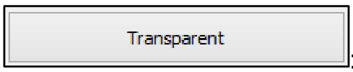
3.4 Common Function

– 

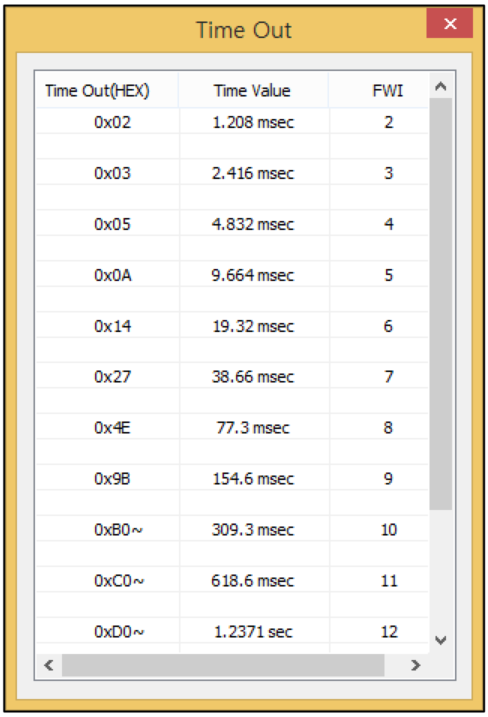
: Input the number of times it takes for the device to send commands and get a response or click the "…" button to automatically generate the value.

– 

: Click to display the "Time-Out" window, <Figure 3-7>.

– 

: Click to transmit a specific protocol or APDU type command that complies with ISO 14443 specifications. Input the PCB, CID and NAD according to your card specification.



Time Out(HEX)	Time Value	FWI
0x02	1.208 msec	2
0x03	2.416 msec	3
0x05	4.832 msec	4
0x0A	9.664 msec	5
0x14	19.32 msec	6
0x27	38.66 msec	7
0x4E	77.3 msec	8
0x9B	154.6 msec	9
0xB0~	309.3 msec	10
0xC0~	618.6 msec	11
0xD0~	1.2371 sec	12

<Figure 3-7>

Time-Out Setting

DualCardDll API :

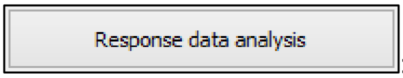
Type A : DEA_Transparent([PORT],[DATA LEN],[DATA],[TIMEOUT],[RLEN],[RBUF]);

Type B : DEB_Transparent([PORT],[DATA LEN],[DATA],[TIMEOUT],[RLEN],[RBUF]);

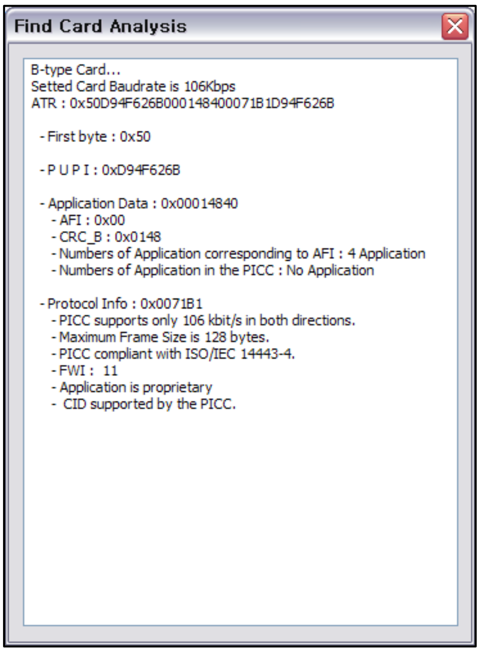
DUALi RW Protocol :

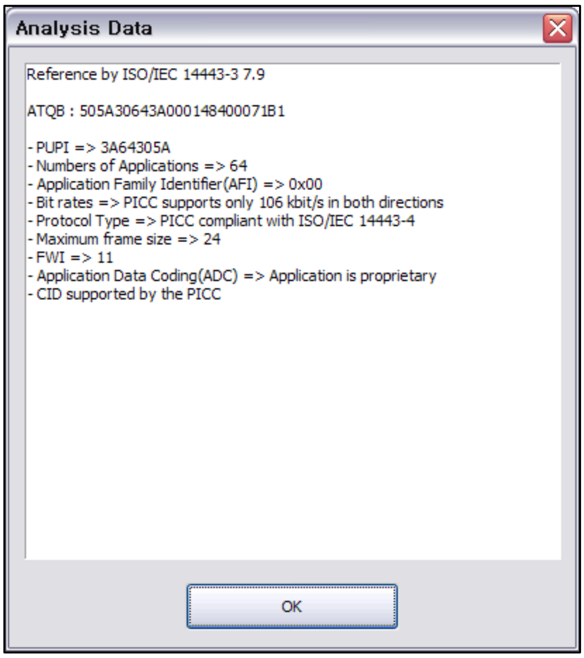
Type A : 41[DATA][TIMEOUT]

Type B : 60[DATA][TIMEOUT]

– 

: Click after executing the [REQA], [Anti-collision], [Select], [RATS], [REQB], or [Detect Card] commands. The response data analysis window will be displayed, as shown in <Figure 3-8> and <Figure 3-9>.





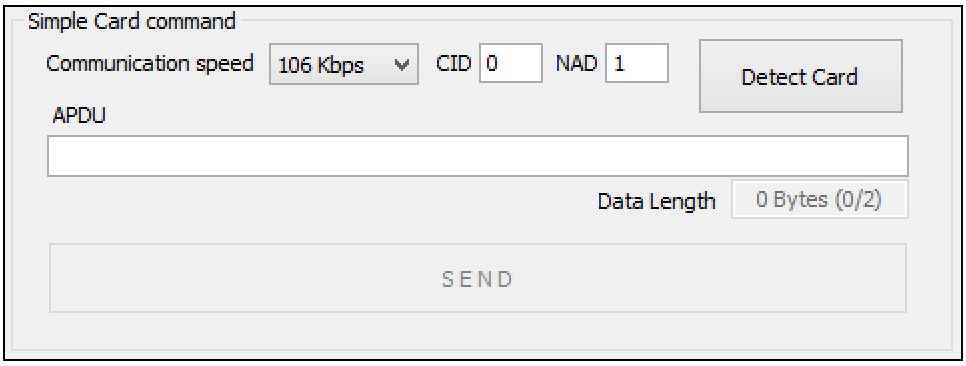
<Figure 3-8>

Analysis for [Detect Card]

<Figure 3-9>

Analysis for [RATS]

3.5 Simple Card Command

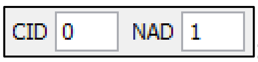


<Figure 3-10>

Simple Card Command

– 

: Select the communication speed, 106/212/424/848 kbps.

– 

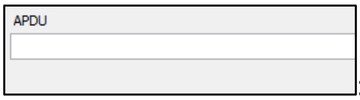
: Set the CID and NAD values for card communication.

– 

: Click to execute the “Detect Card” command.

DualCardDll API : DE_FindCard([PORT],[BAUD],[CID],[NAD],[OPTION],[RLEN],[RBUF]);

DUALi RW Protocol : 4C[BAUD][CID][NAD][OPTION]

– 

: Input the APDU. The PCB and CID are automatically set with the default value (0) and added into the user ulcommand (APDU).

– 

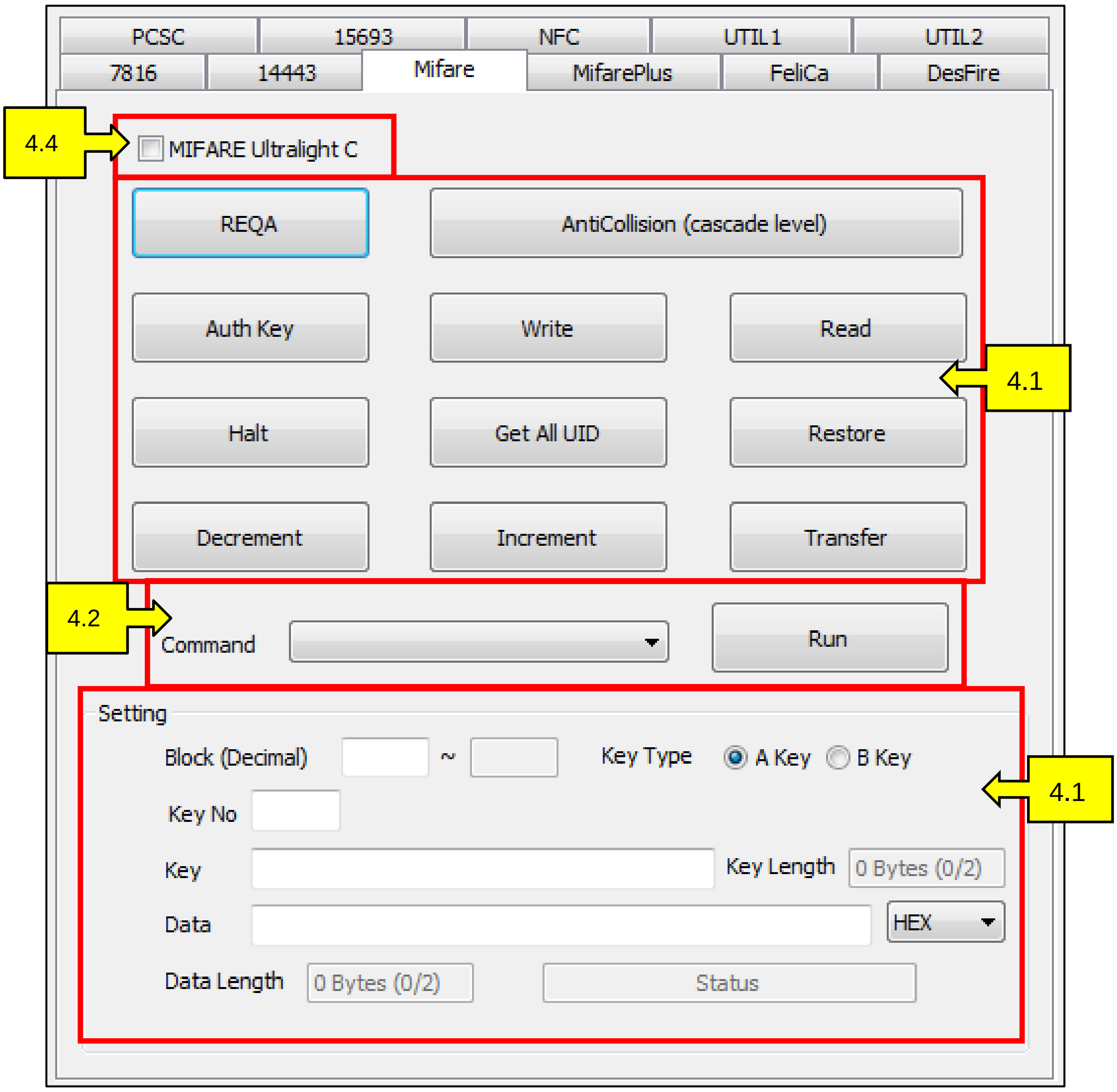
: Click to transmit the APDU

DualCardDll API : DE_APDU([PORT],[SLEN],[SBUF],[RLEN],[RBUF]);

DUALi RW Protocol : 61[DATA]

4. Mifare

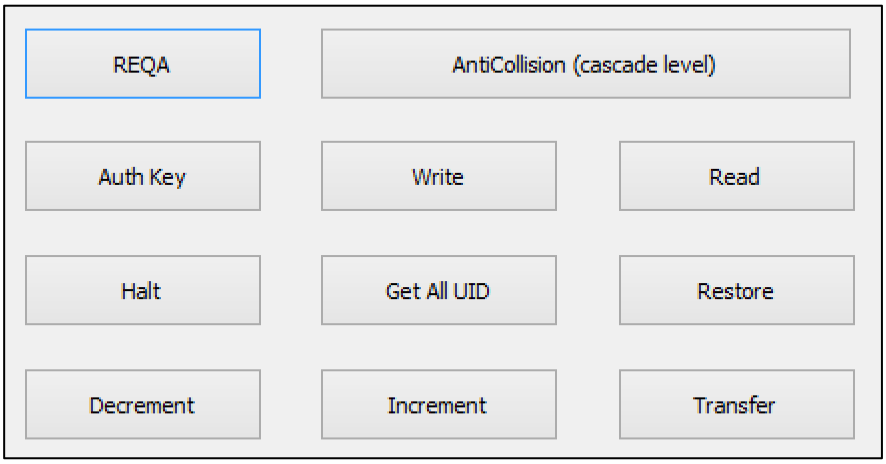
The Mifare tab, <Figure 4-1>, supports communication with NXP's contactless memory card, Mifare. This tab allows users to test all functions in the Mifare specification. Mifare cards (up to layer 2) use the same protocol as ISO 14443 Type A and use their own protocol from layer 3. Mifare specifications require an authentication process to access the memory of Mifare cards. By providing a 6 byte key the system gives permission to read or write data inside the memory. Each block is made of 16 bytes and the data process has to follow the same 16 byte format. Refer to NXP's specification for detailed technical information.



<Figure 4-1>

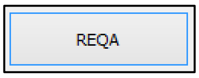
Mifare Tab

4.1 Button Type Command



<Figure 4-2>

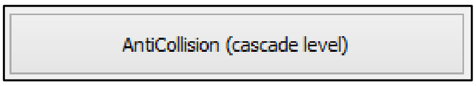
Button Type Command

– 

: Click to execute a “Request” command. A response will be displayed in the log, <Figure 1-7>.

DualCardDll API : DEA_Idle_Req([PORT],[RLEN],[RBUF]);

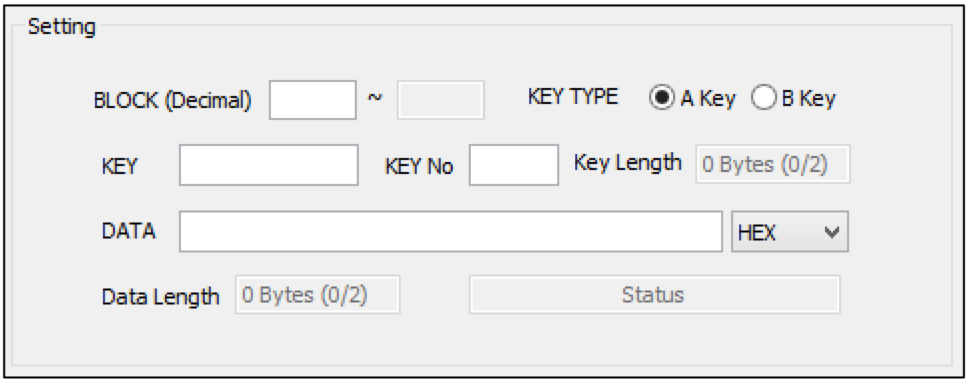
DUALi RW Protocol : 21



: Click to execute an “AntiCollision” or “Select” command. A response will be displayed in the log, <Figure 1-7>.

DualCardDll API : DEA_AntiSelLevel([PORT],[RLEN],[RBUF]);

DUALi RW Protocol : 3D



<Figure 4-3>

Mifare Setting

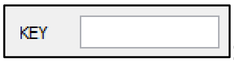
Steps for the Authentication process. **(See Example 14.2**)

1. 

: Input the block number

2. 

: Select a key type for authentication

3. 

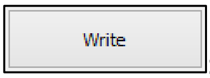
: Input the key value to access a selected block (Hexadecimal, 6byte)

: Click to execute an authorization command. A response will be displayed in the log, <Figure 1-7>.

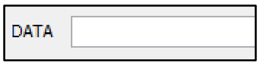
DualCardDll API : DEA_Authkey([PORT],[MODE],[KEY],[BLOCK NO]);

DUALi RW Protocol : 30[MODE][KEY][BLOCK NO]

The following commands must be executed after the Authentication process.

– 

: Click to execute a "Write" command.

– 

: Input data on <Figure 4-3> and click "Write"

DualCardDll API : DEA_Write([PORT],[BLOCK NO],[SLEN],[SBUF]);

DUALi RW Protocol : 28[BLOCK NO][DATA]

– 

: Click to execute a "Read" command

– 

: Input the block number, in <Figure 4-3>, and click "Read" to get a response.

– 

: Click to read a block number. A 16 bytes response will be displayed if the action was successful.

DualCardDll API : DEA_Read([PORT],[BLOCK NO],[RLEN],[RBUF]);

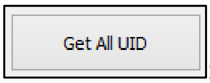
DUALi RW Protocol : 27[BLOCK NO]

– 

: Click to execute a "Halt" command to stop responses. The Mifare card will not respond until it receives a "Wakeup" request.

DualCardDll API : DEA_Halt([PORT]);

DUALi RW Protocol : 26

– 

: Click to read the UID of a card in the RF field.

DualCardDll API : DEA_Req_Select([PORT],[MODE],[RLEN],[RBUF]);

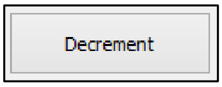
DUALi RW Protocol : 390

– 

: Click to execute a "Restore" command for the Value block.

DualCardDll API : DEA_Restore([PORT],[BLOCK NO]);

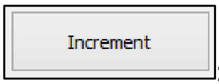
DUALi RW Protocol : 2D[BLOCK NO]

– 

: Click to execute a "Decrement" command for the Value block. (See Example 14.2)

DualCardDll API : DEA_Decrement([PORT],[BLOCK NO],[VALUE]);

DUALi RW Protocol : 2A[BLOCK NO][VALUE]

– 

: Click to execute an "Increment" command for the Value block. (See Example 14.2)

DualCardDll API : DEA_Increment ([PORT],[BLOCK NO],[VALUE]);

DUALi RW Protocol : 29[BLOCK NO][VALUE]

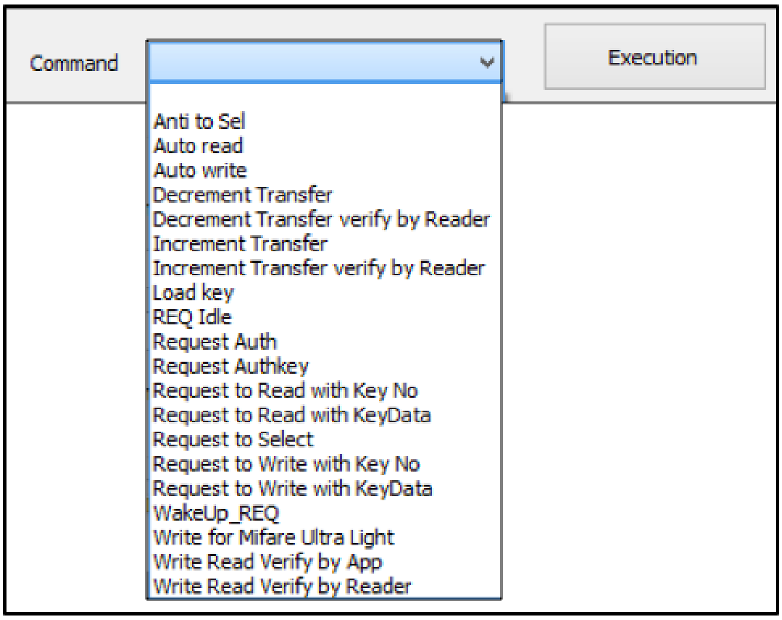
– 

: Select to execute a Transfer command for the Value block.

DualCardDll API : DEA_Transfer([PORT],[BLOCK NO]);

DUALi RW Protocol : 2E[BLOCK NO]

4.2 Combo Box Type Command

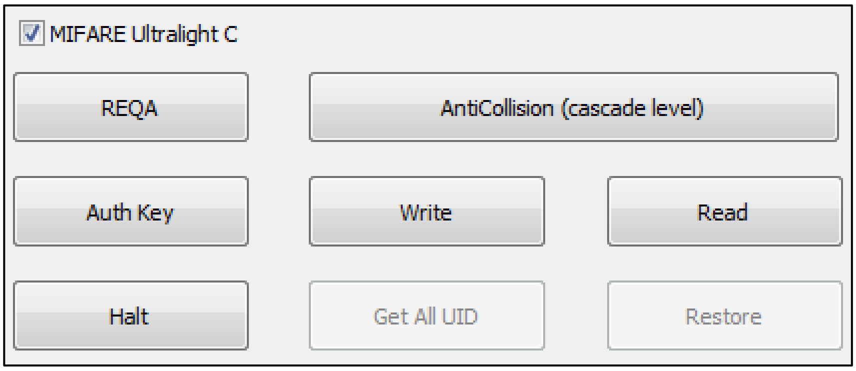


<Figure 4-4>

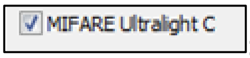
Combo Box Type Command

From the Mifare command list, select a command to execute and click [Execution]. (See Annex 13.1)

4.3 MIFARE Ultralight C



<Figure 4-5>

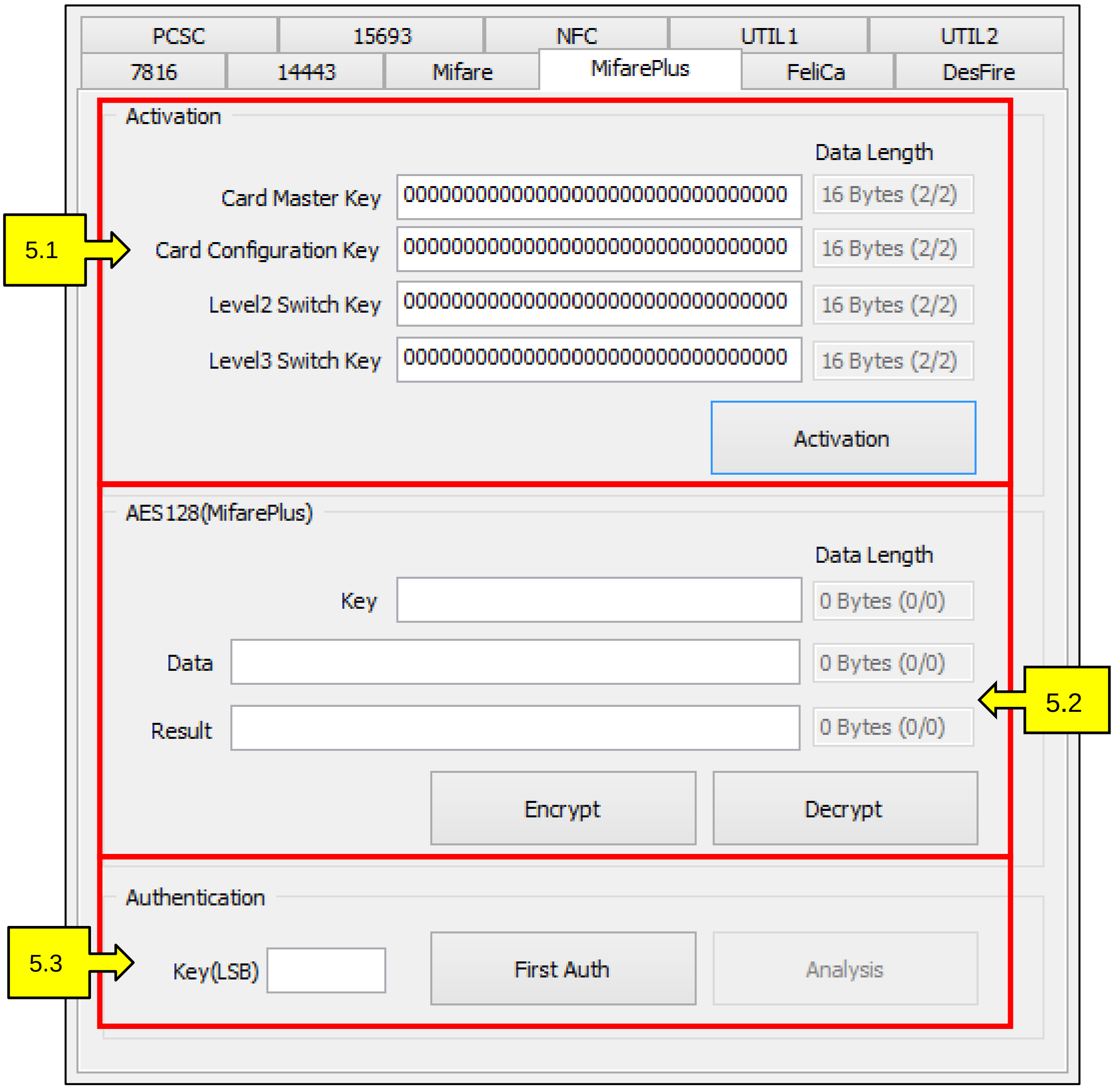
- 

: Click to active MIFARE Ultralight C block. It only activates for [REQA], [AntiCollision], [Auth Key], [Write], [Read], and [Halt] commands.
- After the [Auth Key] is selected, the user can use the Key function.

5. MifarePlus

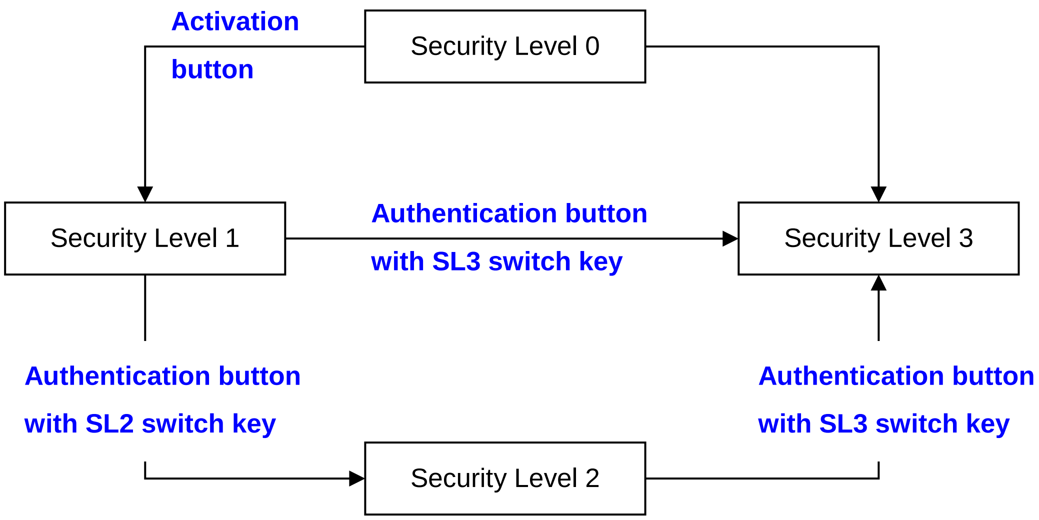
The MifarePlus tab, <Figure 5-1>, supports communication with NXP's MifarePlus. MifarePlus has a more secure function in comparison to Mifare. A unique feature for MifarePlus is that the system will not respond until personalization is conducted.

- MifarePlus level 1 cards are compatible with the Mifare cards.
- MifarePlus level 2 cards use the AES128 algorithm for the authentication process.
- MifarePlus level 3 cards have adopted the ISO 14443 Type A –APDU format.



<Figure 5-1>

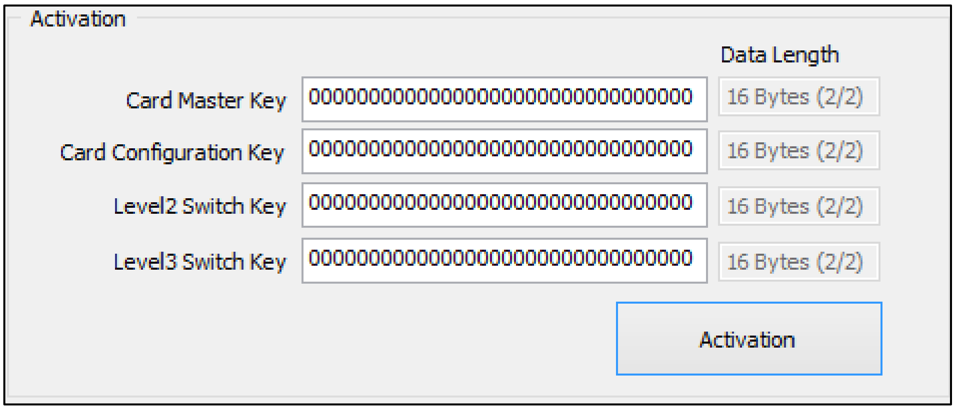
MifarePlus Tab



<Figure 5-2>

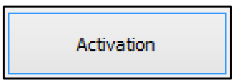
Security Level Switching

5.1 Activation



<Figure 5-3>

MifarePlus Activation



- Click to convert the Security level 0 (default) card to a Security level 1 card. The Security Level 1 card is compatible with the Mifare card.
- Fill out each key (default =0000000…000000) and click “Activation”. It will process as: [REQA] -> [ANTICOLLISION] -> [SELECT] -> [RATS] -> [WRITE PERSO for each key] -> [COMMIT PERSO]
- The activation process sets: [Card Master key], [Card Configuration key], [Level 2 switch key], [Level 3 switch key]. It only sets keys which are mandatory to activate the card. Other keys

can be changed after the activation process and authentication.

- There are “x” & “s” versions of the MifarePlus cards. If using the “x” version, the [Level 2 switch key] is mandatory for activation. However, if using the “s” version, the [Level 2 switch key] is not necessary because the “s” version does not support Level 2.
- After the activation process, the Mifare Plus card becomes a Level 1 card (same as Mifare card). Users can use the “Mifare” tab for the next process (card read & write). Please refer to section 4, Mifare, in the User Manual

DualCardDll API :

REQA : DEA_Idle_Req([PORT],[RLEN],[RBUF]);

ANTICOLLISION, SELECT : DEA_AntiSelLevel([PORT],[RLEN],[RBUF]);

RATS, WRITE PERSO, COMMIT PERSO :

DEA_Transparent([PORT],[SLEN],[SBUF],[TIMEOUT],[RLEN],[RBUF]);

DUALi RW Protocol :

REQA : 21

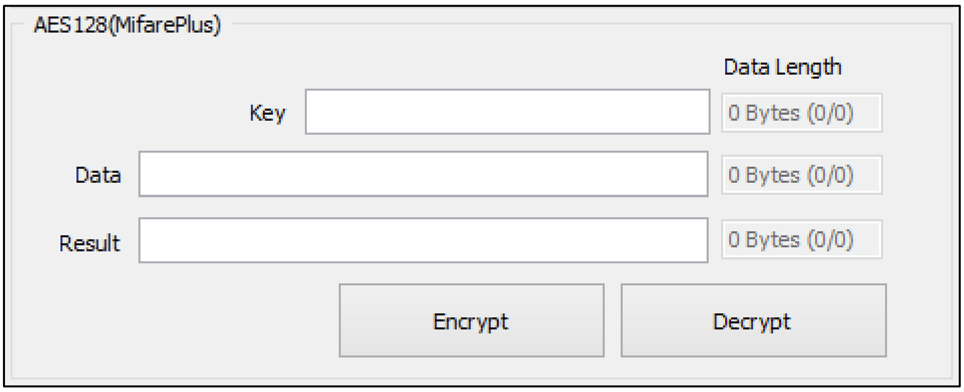
ANTICOLLISION, SELECT : 3D

RATS : 41E080[TIMEOUT]

WRITE PERSO : 41[PCB]A8[KEY NO][KEY][TIMEOUT]

COMMIT PERSO :41[PCB]AA[TIMEOUT]

5.2 AES128



<Figure 5-4>

AES128

- MifarePlus cards use the AES128 algorithm for authentication.
- The AES 128 algorithm is necessary for authentication and communication with Level 2 and Level 3 cards.
- 

: Click after inputting [Key] and [Data]. Results will be displayed in the [Result] field after the data encryption uses the AES128 Algorithm.

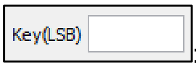
– 

: Click after inputting [Key] and [Data]. Results will be displayed in the [Result] box after the data decryption uses the AES128 Algorithm.

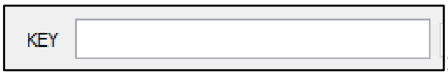
5.3 Authentication

<Figure 5-5>

MifarePlus Authentication

– 

: Input the Key to authenticate.

– 

: Input the Key, in <Figure 5-4>, for selected Key(LSB).

– 

: Click to process the authentication.

– Example: Place a Level 3 card on the DE-620, input the Card master key into [Key] box, and input the key for selected key [0090]. Select [First Auth] to process authentication for the Card master key.

DualCardDll API :

REQA : DEA_Idle_Req([PORT],[RLEN],[RBUF]);

ANTICOLLISION, SELECT : DEA_AntiSelLevel([PORT],[RLEN],[RBUF]);

RATS, FIRST AUTH1, 2 :

DEA_Transparent([PORT],[SLEN],[SBUF],[TIMEOUT],[RLEN],[RBUF]);

DUALi RW Protocol :

REQA : 21

ANTICOLLISION, SELECT : 3D

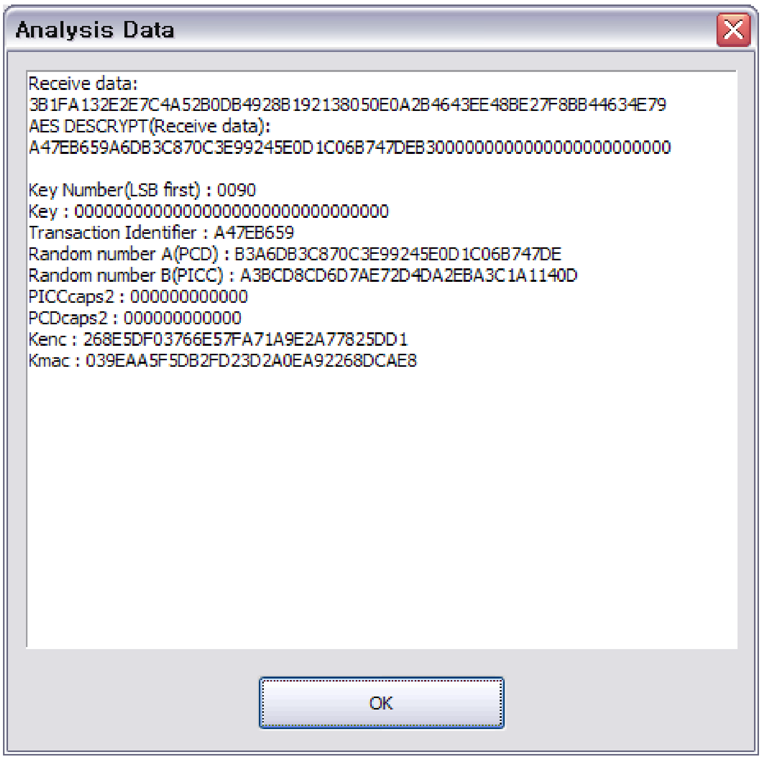
RATS : 41E080[TIMEOUT]

FIRST AUTH1 : 41[PCB]70[KEY NO]06000000000000[TIMEOUT]

FIRST AUTH2 : 41[PCB]72[CIPHER DATA][TIMEOUT]

– 

: Click to see the Analysis Data, <Figure 5-6>. This button will be activated when Authentication is successful.

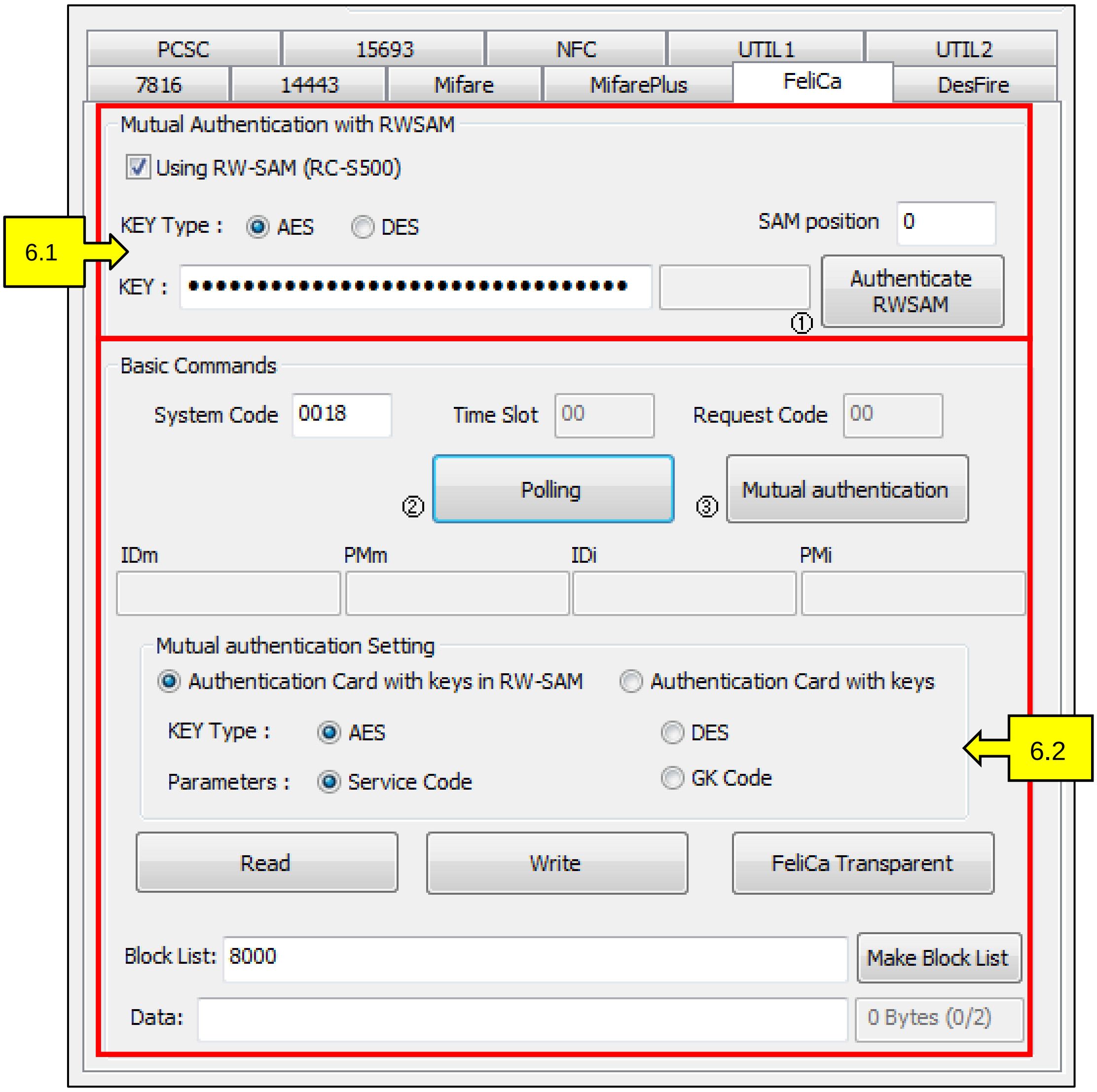


<Figure 5-6>

MifarePlus Analysis Data

6. FeliCa

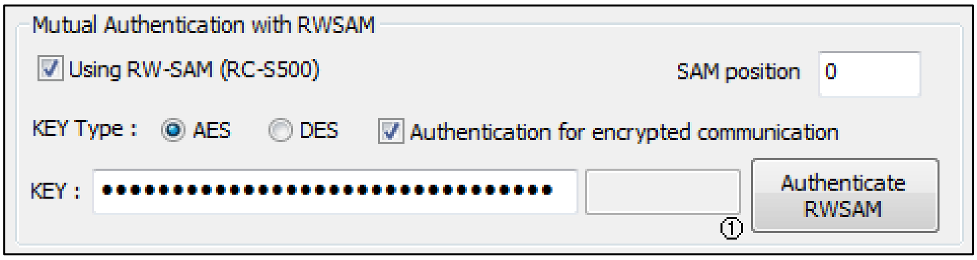
The FeliCa tab, <Figure 6-1>, supports communication with Sony's FeliCa cards. FeliCa cards use the sim types, RW-SAM and RC-S500, which allow higher security during transactions. This program is based on SONY's SAMPLE 5 cards for secured logic and keys. Refer to Sony's FeliCa specification for detailed technology information.



<Figure 6-1>

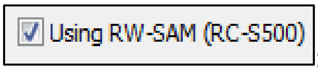
FeliCa Tab

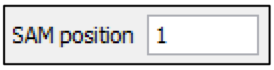
6.1 Mutual Authentication with RW-SAM

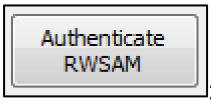


<Figure 6-2>

RWSAM

- 

: Check the box to use encrypted communication with RC-S500 SAM. (For encrypted communication use the RW-SAM).
- 

: Input the SAM slot position.
- 

: Click to execute a mutual authentication. A response will be displayed.
- (This is a batch function for mutual authentications)

  DualCardDll API :

  DE_FeliCa_RWSAMMutualAuthInDES ([PORT], [SLOTNUM] ,[ENCMODE], [KEY], [RLEN], [RBUF])

  DE_FeliCa_RWSAMMutualAuthInAES ([PORT], [SLOTNUM] ,[ENCMODE], [KEY], [RLEN], [RBUF])

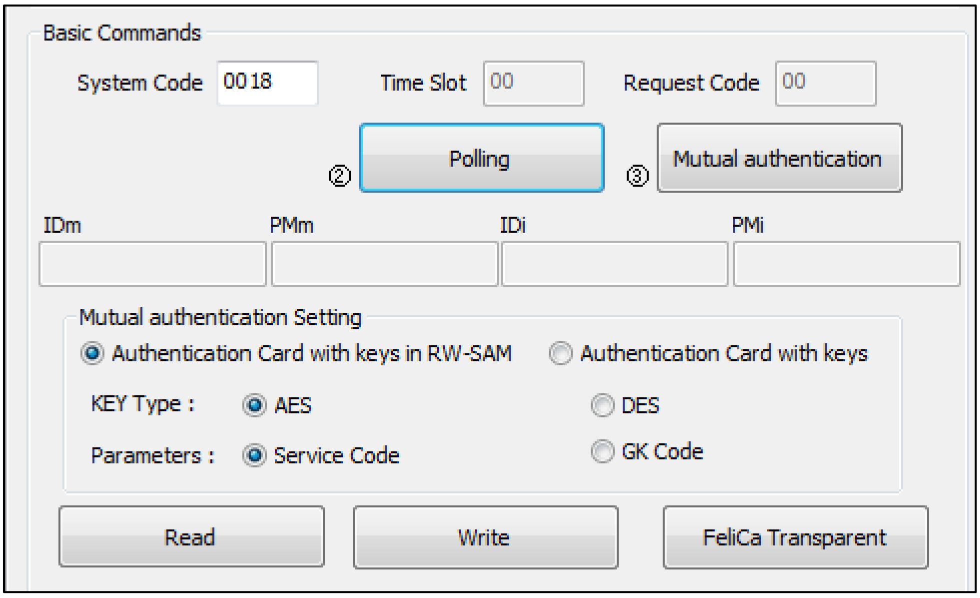
  DUALi RW Protocol :

  5A00[SLOT NO] ,[ENCMODE], [KEY]

  5A01[SLOT NO] ,[ENCMODE], [KEY]
- 

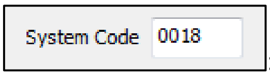
: Select the authentication Key Type.

6.2 FeliCa Communication using RW-SAM

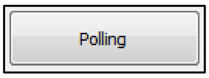


<Figure 6-3>

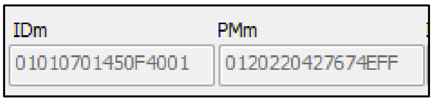
Communication Using RW-SAM

– 

: Input the system code before sending “Polling” commands.

– 

: Click to send “Polling” commands, which come from RW-SAM. A response will be displayed for IDm and PMm. The time slot default value is 00(zero).

– 

: A response from the card will be displayed as [IDm] and [PMm] after a “Polling” command is executed.

DualCardDll API :

DE_FeliCa_Polling ([PORT], [SYSTEMCODE], [RLEN], [RBUF])

DE_FeliCa_PollingWithoutRWSAM ([PORT], [SYSTEMCODE] ,[TIMESLOT], [REQUESTCODE], [RLEN], [RBUF])

DUALi RW Protocol :

5A10[SYSTEMCODE]

5B00[SYSTEMCODE] [TIMESLOT] ,[REQUESTCODE]

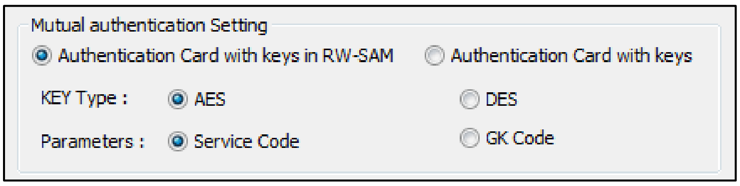
– 

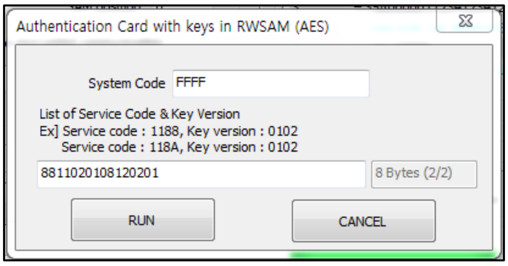
: Click for a Mutual authentication between the RW-SAM and FeliCa card.

Six Modes of Mutual Authentication

The modes depend on the authenticated key type and path for the saving key.

1. Use AES key. Be aware of the system code and key version, select the options below.





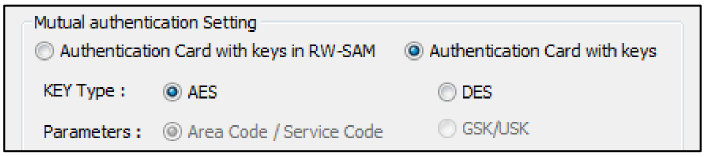
DualCardDll API :

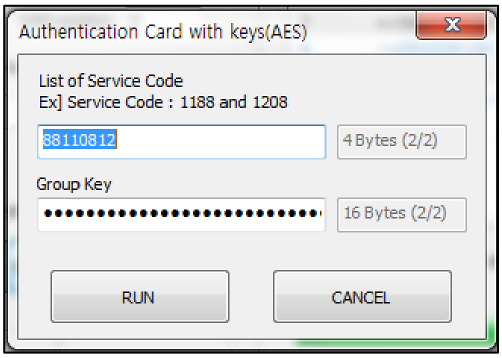
DE_FeliCa_MutualAuthRWSAMInAES_SC ([PORT], [SYSTEMCODE], [nSERVICE], [SERVICECODE&VERSION LIST], [RLEN], [RBUF])

DUALi RW Protocol :

5A06[SYSTEMCODE], [nSERVICE], [SERVICECODE&VERSION LIST]

2. Use AES key. Be aware of the service code and Group key. Select the options below.





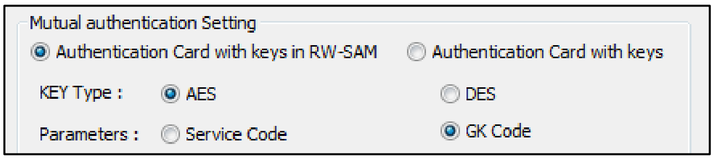
DualCardDll API :

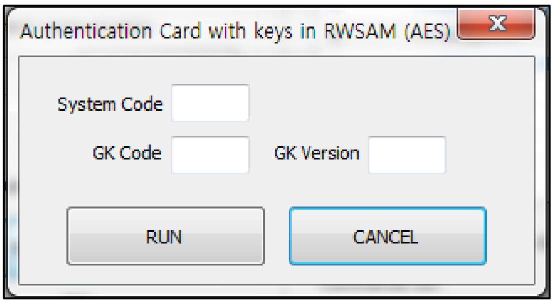
DE_FeliCa_CardMutualAuthInAES_GK ([PORT], [nSERVICE], [SERVICECODE LIST],[GROUP KEY], [RLEN], [RBUF])

DUALi RW Protocol :

5A05 [nSERVICE], [SERVICECODE LIST],[GROUP KEY]

3. Use AES key. Be aware of system code , Group key code, and version, select the options below.





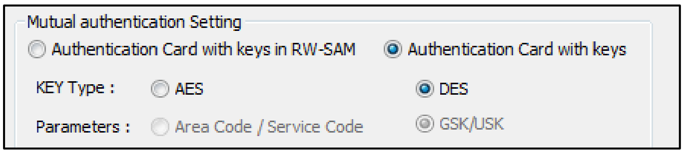
DualCardDll API :

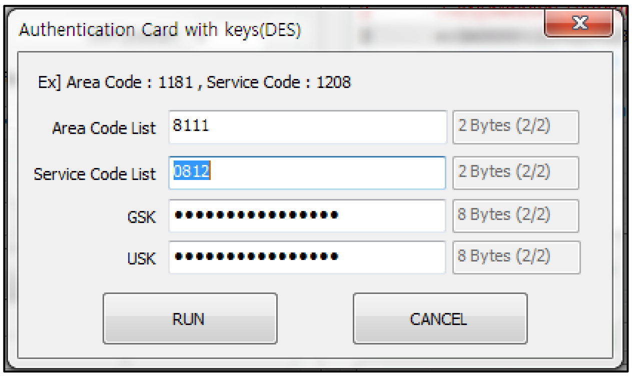
DE_FeliCa_MutualAuthRWSAMInAES_GKC ([PORT], [SYSTEMCODE], [GROUP KEY CODE], [GROUP KEY VERSION], [RLEN], [RBUF])

DUALi RW Protocol :

5A07[SYSTEMCODE], [GROUP KEY CODE], [GROUP KEY VERSION]

4. Use DES key. Be aware of Area code, Service code, Group service key, User service key. Select the options below.





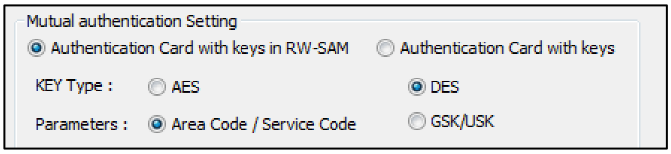
DualCardDll API :

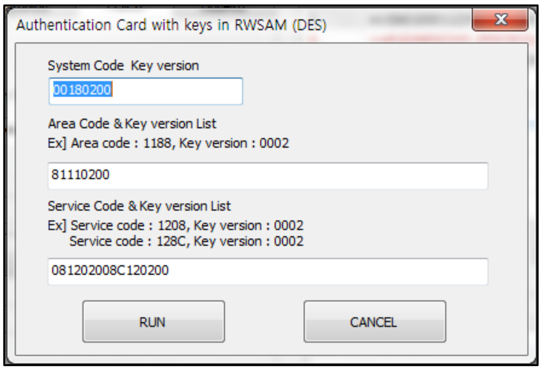
DE_FeliCa_CardMutualAuthInDES_GSKUSK ([PORT], [nAREA], [AREACODE LIST], [nSERVICE], [SERVICECODE LIST],[GROUP SERVICE KEY], [USER SERVICE KEY], [RLEN], [RBUF])

DUALi RW Protocol :

5A02[nAREA], [AREACODE LIST], [nSERVICE], [SERVICECODE LIST],[GROUP SERVICE KEY], [USER SERVICE KEY]

5. Use DES key. Be aware of the Area code & version, Service code & version. Select the options below.





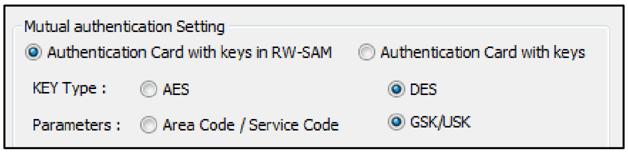
DualCardDll API :

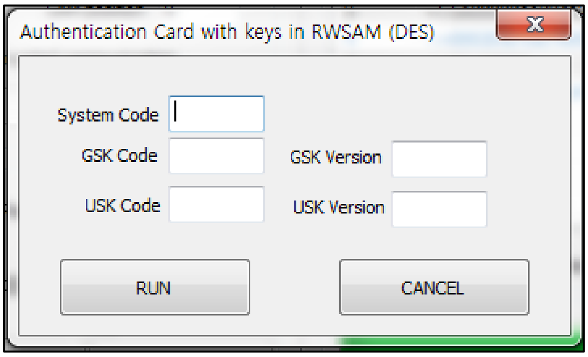
DE_FeliCa_MutualAuthRWSAMInDES_ACSC ([PORT], [SYSTEMCODE], [SYSTEMKEY VERSION], [nAREA], [AREACODE&VERSION LIST], [nSERVICE], [SERVICECODE&VERSION LIST], [RLEN], [RBUF])

DUALi RW Protocol :

5A04[SYSTEMCODE], [SYSTEMKEY VERSION], [nAREA], [AREACODE&VERSION LIST], [nSERVICE], [SERVICECODE&VERSION LIST]

6. Use DES key. Be aware of Group service key & version, User service key & version. Select the options below.





DualCardDll API :

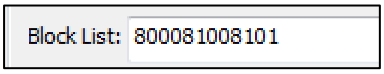
DE_FeliCa_MutualAuthRWSAMInDES_GSKUSKVersion ([PORT], [SYSTEMCODE], [GSKCODE], [GSKVERSION], [USKCODE],[USKVERSION],[RLEN], [RBUF])

DUALi RW Protocol :

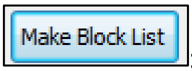
5A03[SYSTEMCODE], [GSKCODE], [GSKVERSION], [USKCODE],[USKVERSION]

– 

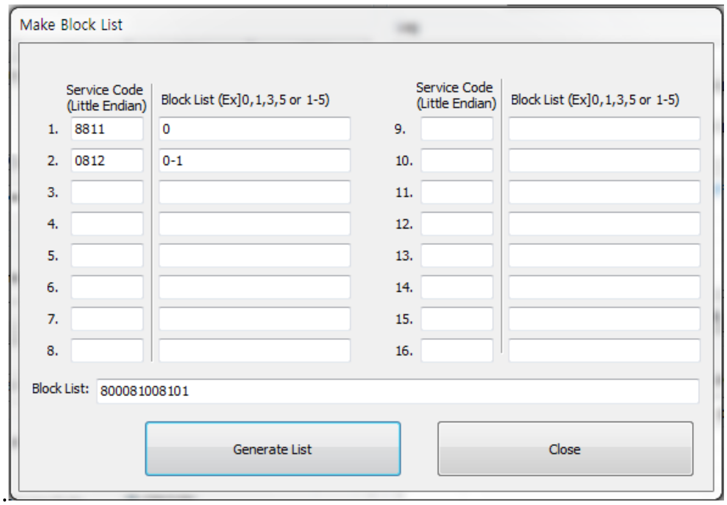
: Click to get data from the input block list.

– 

: Input [Block list] directly.

– 

: Click to create a block list with information of service codes and block numbers. Service codes. should be in the same order as when you used the mutual authentication.



<Figure 6-4>

Block List

DualCardDll API :

DE_FeliCa_ReadBlock([PORT], [nBLOCK], [BLOCKLIST],[RLEN], [RBUF])

DE_FeliCa_ReadBlockWithoutEnc([PORT], [nSERVICE], [SERVICECODELIST], [nBLOCK], [BLOCKLIST], [RLEN], [RBUF])

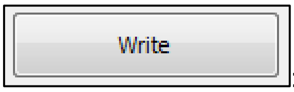
DE_FeliCa_ReadBlockWithoutEnc([PORT], [nSERVICE], [SERVICECODELIST], [nBLOCK], [BLOCKLIST], [RLEN], [RBUF])

DUALi RW Protocol :

5A11[nBLOCK], [BLOCKLIST]

5A13[nSERVICE], [SERVICECODELIST], [nBLOCK], [BLOCKLIST]

5B01[nSERVICE], [SERVICECODELIST], [nBLOCK], [BLOCKLIST]

- 

: Click to write data of input block list.

DualCardDll API :

DE_FeliCa_WriteBlock([PORT], [nBLOCK], [BLOCKLIST], [DATA], [RLEN], [RBUF])

DE_FeliCa_WriteBlockWithoutEnc([PORT], [nSERVICE], [SERVICECODELIST], [nBLOCK], [BLOCKLIST], [DATA], [RLEN], [RBUF])

DE_FeliCa_WriteBlockWithoutEnc([PORT], [nSERVICE], [SERVICECODELIST], [nBLOCK], [BLOCKLIST], [DATA], [RLEN], [RBUF])

DUALi RW Protocol :

5A12[nBLOCK], [BLOCKLIST],[DATA]

5A14[nSERVICE], [SERVICECODELIST], [nBLOCK], [BLOCKLIST], [DATA]

5B02[nSERVICE], [SERVICECODELIST], [nBLOCK], [BLOCKLIST], [DATA]

- 

: Click to send commands to Felica cards.

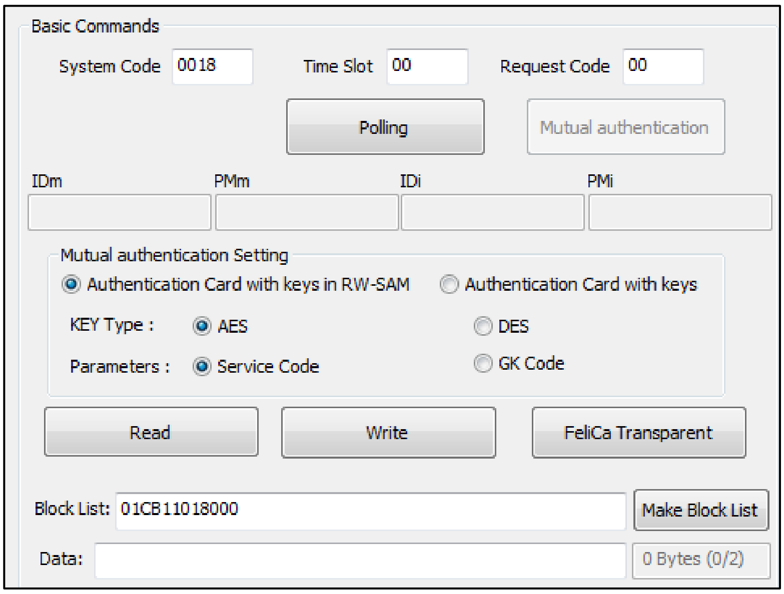
DualCardDll API :

DEC_Transparent ([PORT],[DATA LEN],[DATA],[RLEN],[RBUF],[TIMEOUT]);

DUALi RW Protocol :

50[DATA LEN],[DATA] ,[TIMEOUT]

6.3 FeliCa Communication NOT using RW-SAM



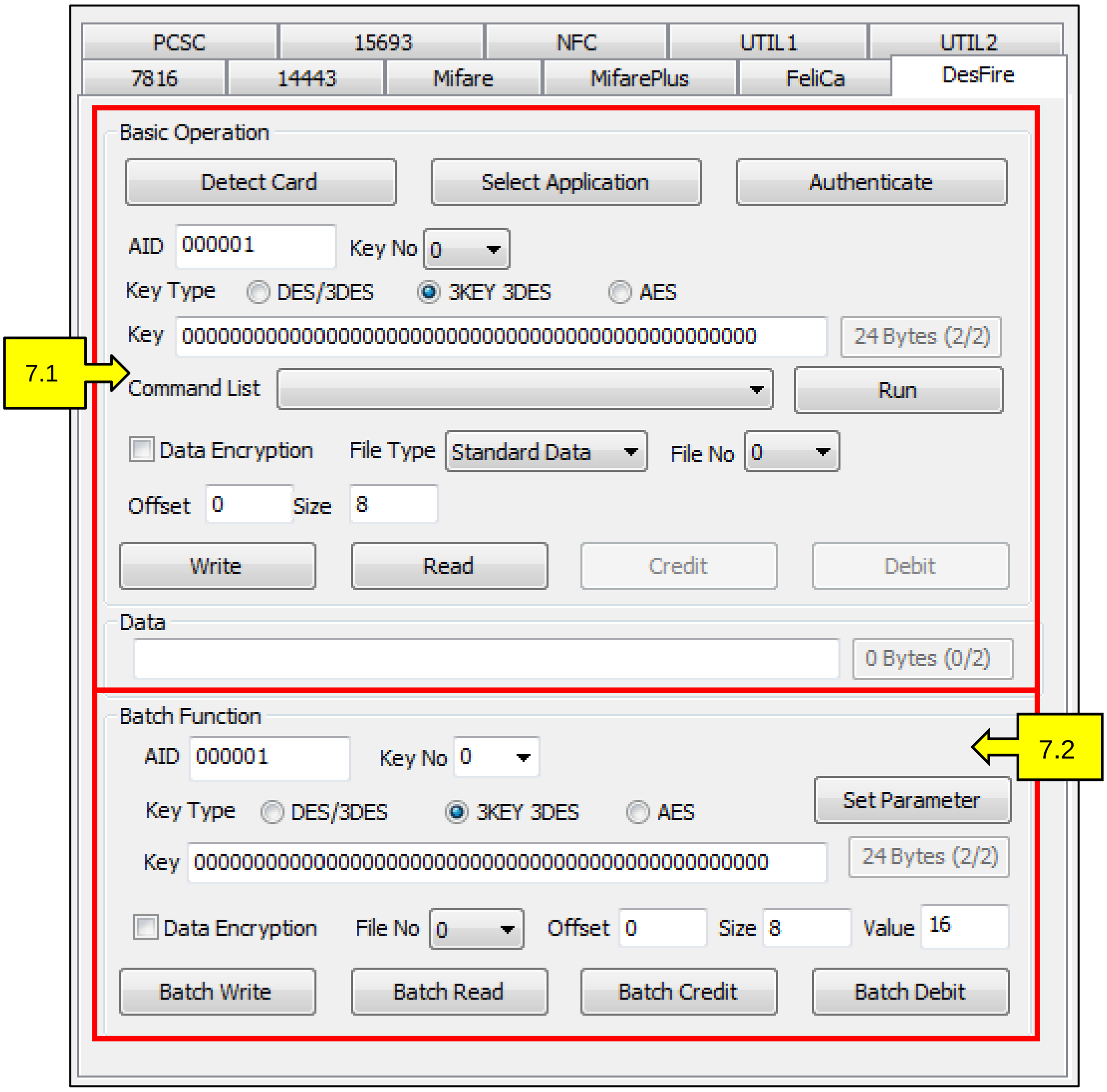
<Figure 6-5>

Communication Not Using RW-SAM

- After “Polling”, you can skip “Mutual Authentication” and proceed to the read/write function

7. DesFire

The DesFire tab, <Figure 7-1>, supports communication with NXP's DesFire cards.

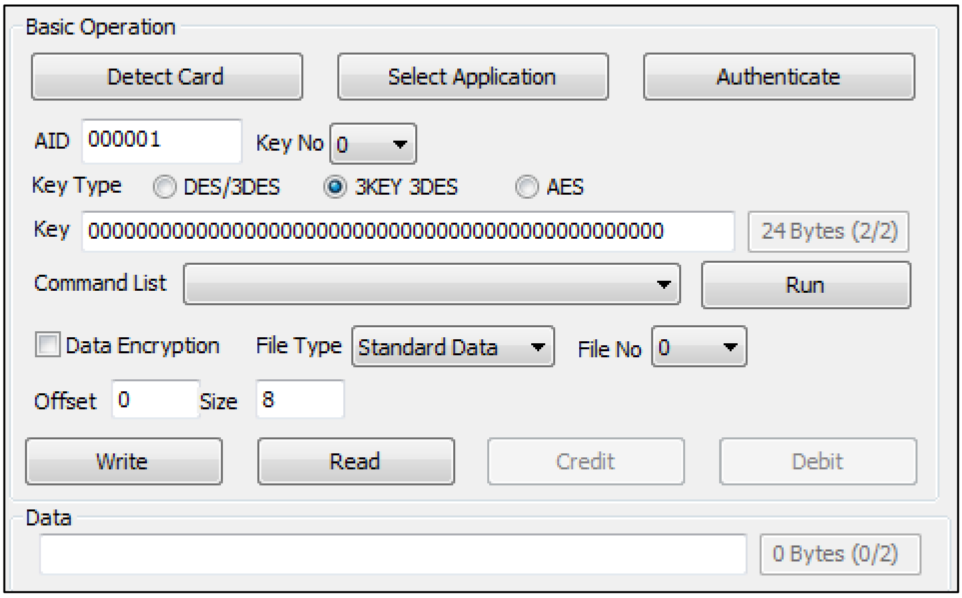


<Figure 7-1>

DesFire Tab

7.1 Basic Operation

To use DesFire cards, first select the "AID" application then authenticate.



<Figure 7-2>

DesFire Basic Operation

– 

: Click to confirm that the connected device is detecting cards

DualCardDll API :

DE_FindCard([PORT],[BAUD],[CID],[NID],[OPTION],[RLEN],[RBUF]);

DUALi RW Protocol :

4C00000000

– 

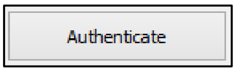
: Select to choose an application ID

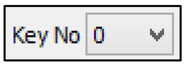
DualCardDll API :

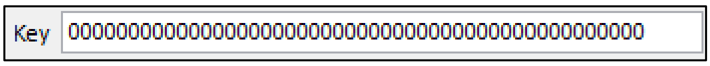
DE_DESFireTransparent([PORT],[FALG],[CMD LEN],[SLEN],[SBUF],[RLEN],[RBUF]);

DUALi RW Protocol :

(RW_Protocol_DesFire(150109) Refer to Chapter 5.1) 3A0200045A[AID]

– 

: Click after inputting the corresponding 

key number, key data 

, and AID (application ID) to authenticate. When the application ID is set to "000000", the user will have full control of the PICC level.

DualCardDll API :

DE_DESFireAuthentication([PORT],[KEY NO],[KEY],[KEY LEN],[RBUF],[RLEN]);

DE_DESFireAuthentication_AES([PORT],[KEY NO],[KEY],[KEY LEN],[RBUF],[RLEN]);

DUALi RW Protocol :

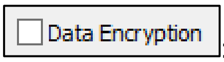
(Refer to RW_Protocol sepc (150109) Chapter 5.1 DesFire)

3A01[KEY NO][KEY]

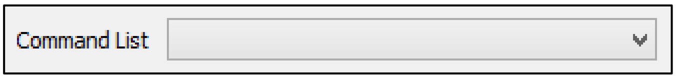
3A05[KEY NO][KEY]

– 

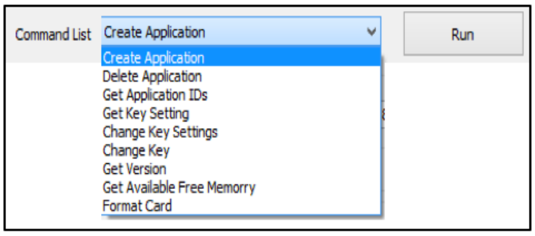
: Select Key Type.

– 

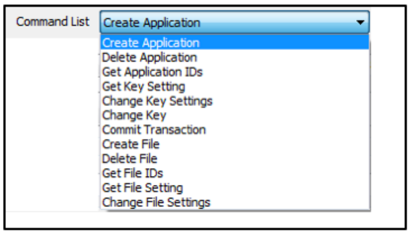
: Check the box to encrypt commands (file level command)

– 

: Select the appropriate command set. Different command lists will appear in the dropdown box when you select PICC level and the application level, as shown in <Figure 7.3> and <Figure 7.4>.



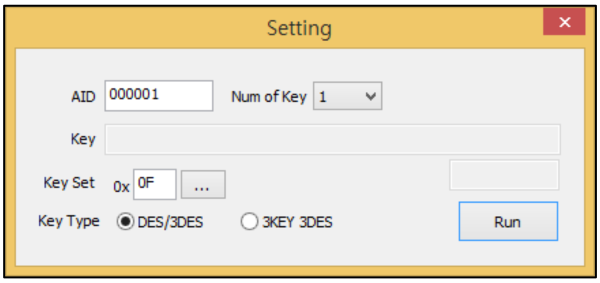
<Figure 7-3 Command for PICC Level>



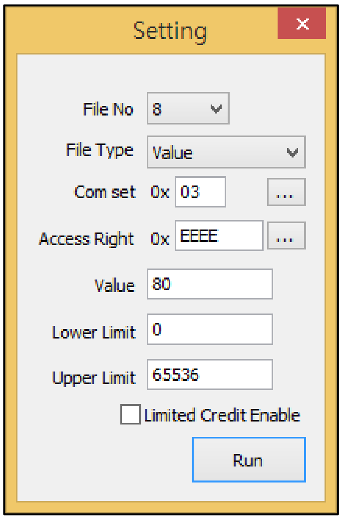
<Figure 7-4 Command for App Level>

– 

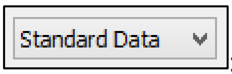
: Click to execute the selected command in the command list. When additional information is required, the windows, <Figure 7-5> and <Figure 7-6>, will pop up.



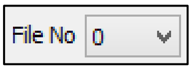
<Figure 7-5 Setting for Keys>



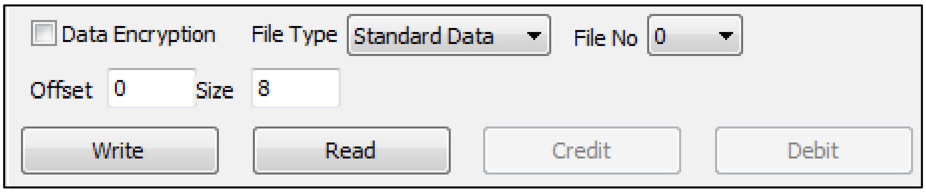
<Figure 7-6 Setting for Files>

– 

: Select a file type. Depending on the selected “File Type”, different setting options will appear in the group box.

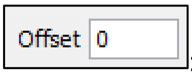
– 

: Select the file number.

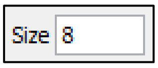


<Figure 7-7>

File Type, Standard Data

– 

: Input the offset to read and write files.

– 

: Input the size to read and write files.

– 

: Click to write files.

DualCardDll API:

DE_DESFireTransparent([PORT],[FALG],[CMD LEN],[SLEN],[SBUF],[RLEN],[RBUF]);

DUALi RW Protocol:

3A02[FLAG]083D[FILE NO][OFFSET][FILE SIZE][DATA] data

3A02[FLAG]083B[FILE NO][OFFSET][FILE SIZE][DATA] record

3A02[FLAG]01AF

– 

: Click to read files.

DualCardDll API :

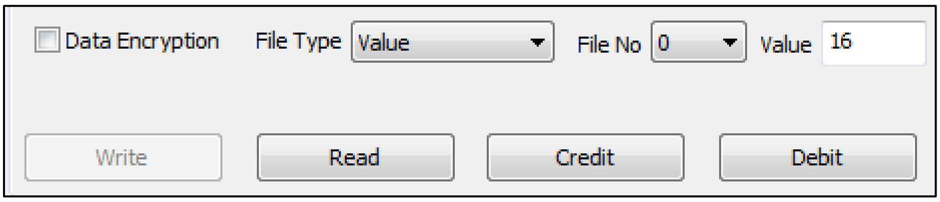
DE_DESFireTransparent([PORT],[FALG],[CMD LEN],[SLEN],[SBUF],[RLEN],[RBUF]);

DUALi RW Protocol :

3A02[FLAG]08BD[FILE NO][OFFSET][FILE SIZE] data

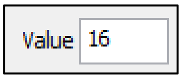
3A02[FLAG]08BB[FILE NO][OFFSET][FILE SIZE] record

3A02[FLAG]01AF

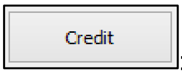


<Figure 7-8>

File Type, Value

– 

: Input value of "Credit" or "Debit".

– 

: Click to add the number of credits, listed in the "Value" field.

DualCardDll API :

DE_DESFireTransparent([PORT],[FALG],[CMD LEN],[SLEN],[SBUF],[RLEN],[RBUF]);

DUALi RW Protocol :

3A02[FLAG]020C[FILE NO][VALUE]

– 

: Click to deduct the number of credits, listed in the "Value" field.

DualCardDll API :

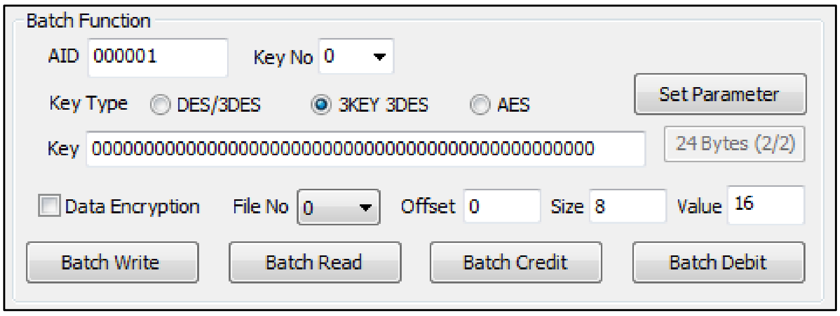
DE_DESFireTransparent([PORT],[FALG],[CMD LEN],[SLEN],[SBUF],[RLEN],[RBUF]);

DUALi RW Protocol :

3A02[FLAG]02DC[FILE NO][VALUE]

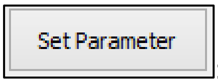
7.2 Batch Command

Using the "Batch" function is a faster and more convenient to execute mulitple commands, especially when the user is not familiar with Desfire speciications and structure. Dualcard performs batch commands that shortcut the process from "Detect" to the "selected command protocol".



<Figure 7-9>

Batch Function

- 

: Click to set the Application ID, Key Number and Key value on the device.

  DualCardDll API :

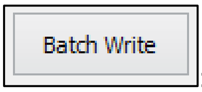
  DE_DESFire_SetConfig_Batch([PORT], [AID], [KEY], [KEY LEN], [KEY NO]);

  DE_DESFire_SetConfig_Batch_AES([PORT], [AID], [KEY], [KEY LEN], [KEY NO]);

  DUALi RW Protocol : (Refer to RW_Protocol sepc (150109) Chapter 5.1 DesFire)

  3A03[AID][KEY][KEY NO]

  3A06[AID][KEY][KEY NO]

- 

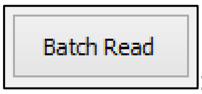
: Click to execute the batch command from “Detect” file to “Write” file

  DualCardDll API :

  DE_DESFire_WriteFile_Batch ([PORT], [FLAG], [FILE NO], [OFFSET], [FILE SIZE], [DATA]);

  DUALi RW Protocol : (Refer to RW_Protocol sepc (150109) Chapter 5.1 DesFire)

  3A04[FALG]3D[FILE NO][OFFSET][FILE SIZE][DATA]

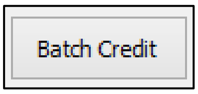
- 

: Click to execute the batch command from “Detect” file to “Read” file

  DualCardDll API : DE_DESFire_ReadFile_Batch ([PORT], [FLAG], [FILE NO], [OFFSET], [FILE SIZE], [DATA LEN], [DATA]);

  DUALi RW Protocol : (Refer to RW_Protocol sepc (150109) Chapter 5.1 DesFire)

  3A04[FALG]BD[FILE NO][OFFSET][FILE SIZE]

- 

: Click to execute the batch command from “Detect” file to “Credit”. This command adds the number of credits, listed in the “Value” field, to the value type. By using the “Batch Credit” option, you are conducting a single step instead of four steps (“Detect Card”, “Select Application”, “Authenticate”, and “Credit”).

  DualCardDll API :

  DE_DESFire_Credit_Batch ([PORT], [FLAG], [FILE NO], [VALUE]);

  DUALi RW Protocol : (Refer to RW_Protocol sepc (150109) Chapter 5.1 DesFire)

  3A04[FALG]0C[FILE NO][OFFSET][FILE SIZE][DATA]

- 

: Click to execute the batch command from “Detect” file to “Debit”. This command subtracts the number of debits, listed in the “Value” field, to the value type. By using the “Batch Debit” option, you are conducting a single step instead of four steps (“Detect Card”, “Select Application”, “Authenticate”, and “Debit”).

  DualCardDll API:

  DE_DESFire_Debit_Batch ([PORT], [FLAG], [FILE NO], [VALUE]);

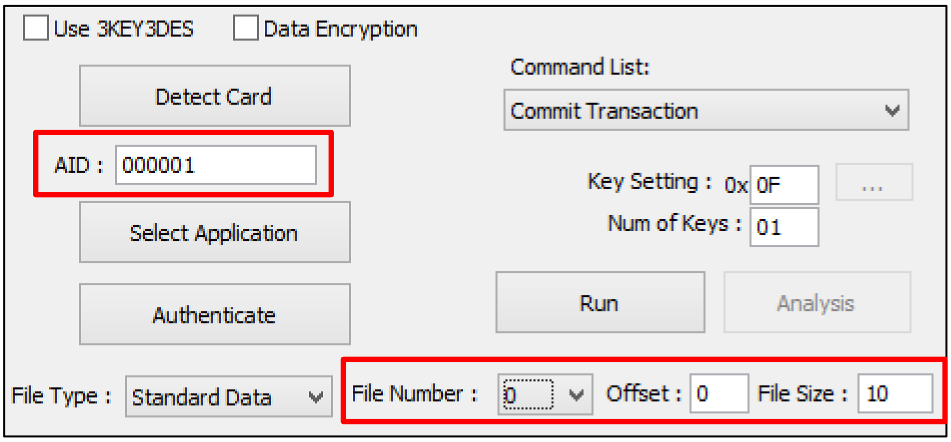
  DUALi RW Protocol: (Refer to RW_Protocol sepc (150109) Chapter 5.1 DesFire)

3A04[FALG]DC[FILE NO][OFFSET][FILE SIZE][DATA]

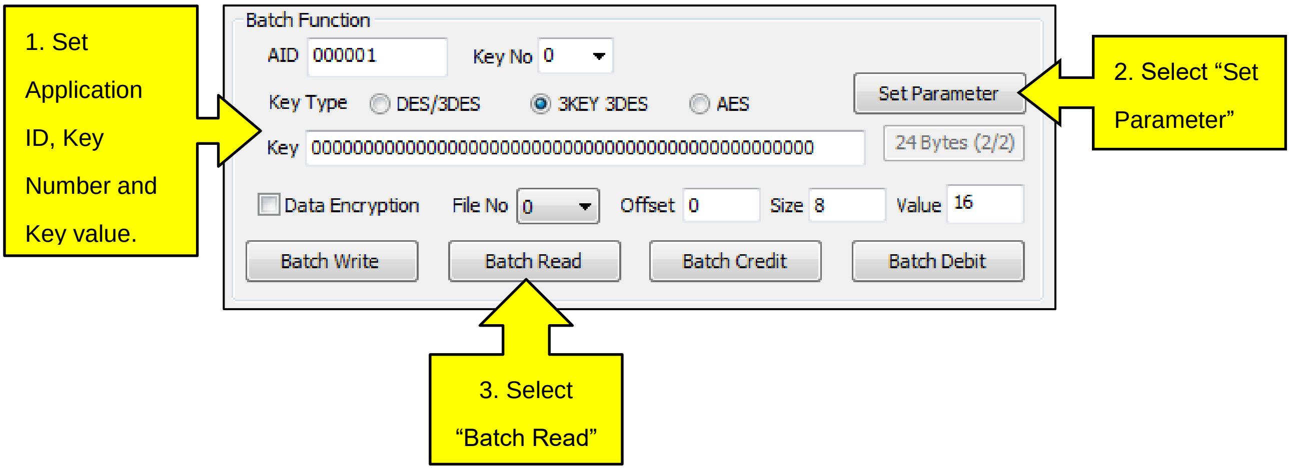
– 

: Select Key Type

Example 14.4.2, Batch Command

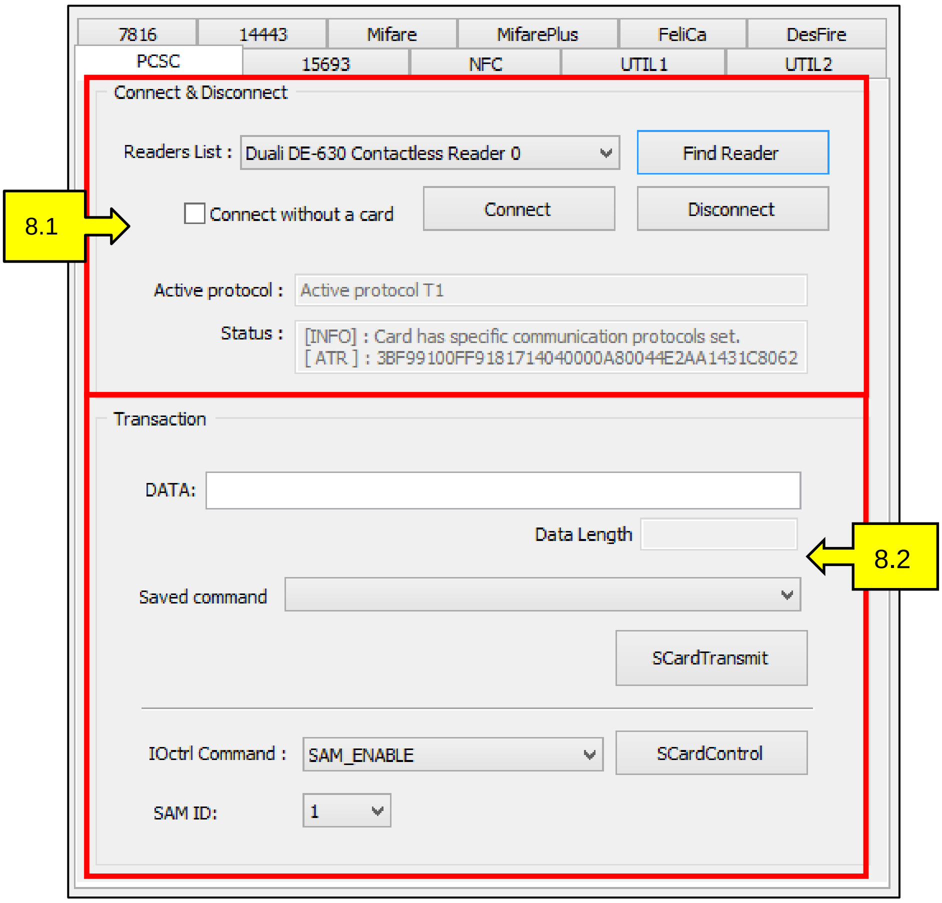


- [Batch Read]



8. PCSC

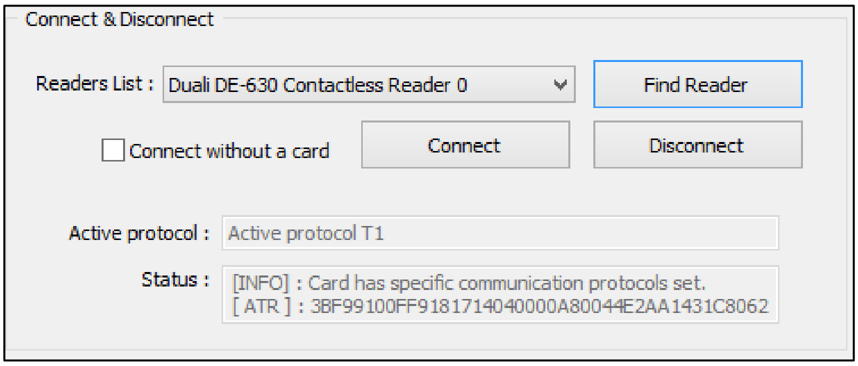
This tab, <Figure 8-1>, supports communication standards from Microsoft Window and PCSC.



<Figure 8-1>

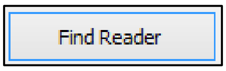
PCSC Tab

8.1 Connect and Disconnect

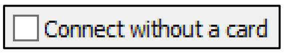


<Figure 8 -2>

PCSC Connect and Disconnect

- 

: Click to find the PCSC device that is connected to the PC.
- 

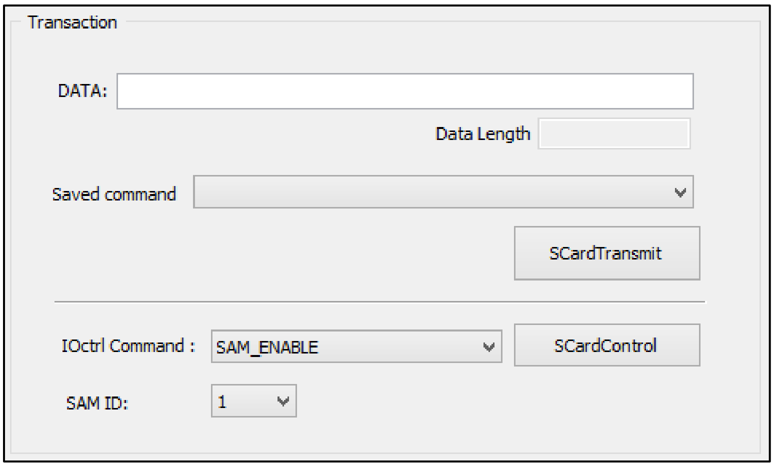
: Select a device that has been detected.
- 

: Check the box if an RF card is not being used during this process.
- 

: Click to connect the selected device. When a card is placed on the device, the "Status" will display the ATR value. If there is no RF card, select "Disconnect".
- 

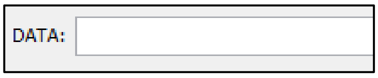
: Click to disconnect the PCSC device.

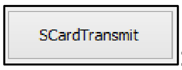
8.2 Transaction



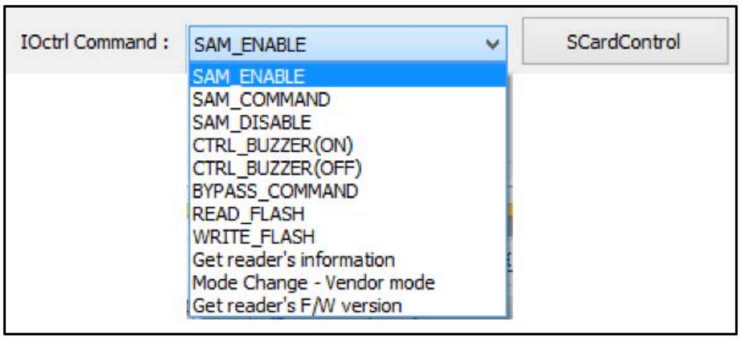
<Figure 8-3>

PCSC Transaction

- 

: Input a command, this is optional.
- 

: Click to execute the data or command in the "Data" field.



<Figure 8-4>

IOctrl Command list

- To use the IOctrl Command from the PCSC communication specification:

: Select the SAM, only if there are more than 2 SAMs

: Click to execute the IOctrl Command.

9. 15693 (Optional)

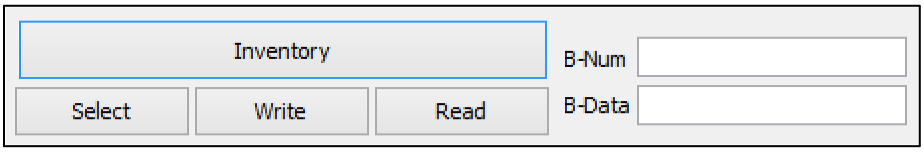
This tab, <Figure 9-1>, supports communication with ISO 15693. Refer to ISO 15693 or ISO 18000-3 specifications for detailed technical information.

*Contact the DUALi Lab to verify if your device supports the ISO 15693 protocol.

<Figure 9-1>

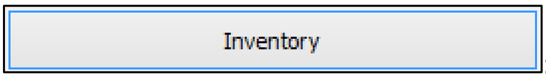
15693 Tab

9.1 Inventory & Select & Write & Read



<Figure 9-2>

15693 Inventory

– 

: Click to execute the inventory command. This command reads multiple card UIDs at one time.

DualCardDll API : DED_Inventory([PORT],[FLAG],[RLEN],[RBUF]);

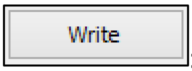
DUALi RW Protocol : 7001

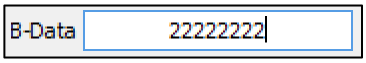
– 

: Click to select the last detected card.

DualCardDll API : DED_Select([PORT],[RLEN],[RBUF]);

DUALi RW Protocol : 71

– 

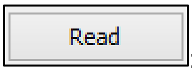
: Click to write the B-Data, 

, in the B-Num, 

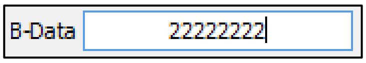
.

DualCardDll API : DED_Write([PORT],[BLOCK NO],[DATA],[RLEN],[RBUF]);

DUALi RW Protocol : 73[BLOCK NO][DATA]

– 

: Click to read the B-Data in the B-Num, 

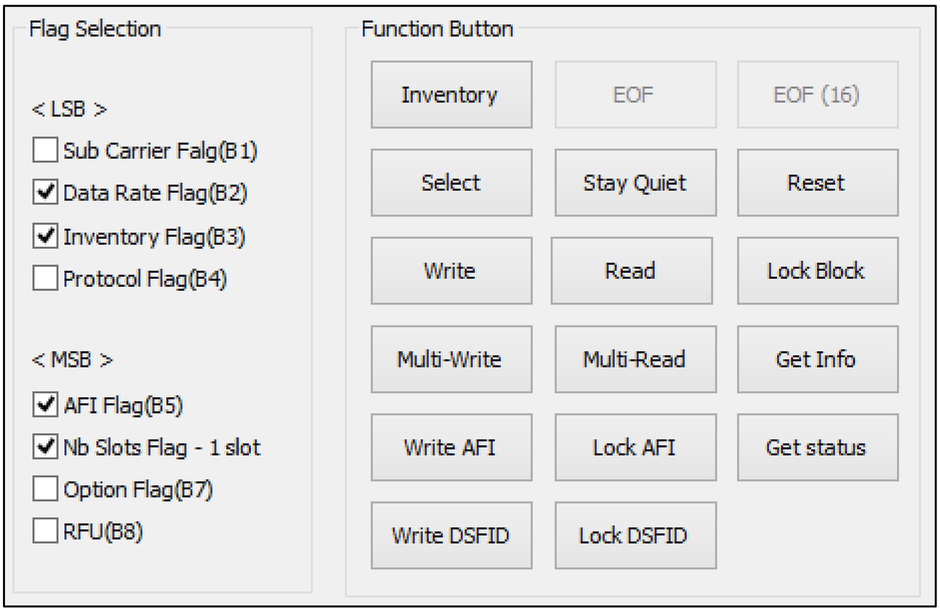
. The B-Data will be displayed in 

.

DualCardDll API : DED_Read([PORT],[BLOCK NO],[RLEN],[RBUF]);

DUALi RW Protocol : 72[BLOCK NO]

9.2 Select Flag & Create Command



<Figure 9-3>

Flag Selection and Function Button

- The “Flag Selection” group box allows users to set flags for when devices send commands to card. Refer to <Figure 9-4>.

- The “he gure 9-4>.dis group box has a list ISO15693 commands. You can also select commands from a list, refer to <Figure 9-4>.

- 

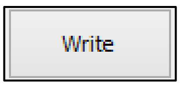
, DUALi RW Protocol : 74[FLAG]01[BLOCK NO][BLOCK COUNT]
- 

, DualCardDll API : DED_Eof([PORT],[RLEN],[RBUF]);
DUALi RW Protocol : 78
- 

, DualCardDll API : DED_Eof([PORT],[RLEN],[RBUF]);
DUALi RW Protocol : 78
- 

, DUALi RW Protocol : 74[FLAG]25[UID]
- 

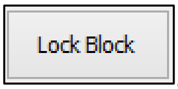
, DUALi RW Protocol : 74[FLAG]02[UID]
- 

, DUALi RW Protocol : 74[FLAG]26[UID]
- 

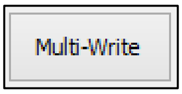
, DUALi RW Protocol :74[FLAG]21[UID][BLOCK NO][BLOCK COUNT]

– 

, DUALi RW Protocol : 74[FLAG]20[UID][BLOCK NO]

– 

, DUALi RW Protocol : 74[FLAG]22[UID][BLOCK NO]

– 

, DUALi RW Protocol : 74[FLAG]24[UID][BLOCK NO][BLOCK COUNT][BLOCK DATA]

– 

, DUALi RW Protocol : 74[FLAG]23[UID][BLOCK NO][BLOCK COUNT]

– 

, DUALi RW Protocol : 74[FLAG]2B[UID]

– 

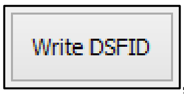
, DUALi RW Protocol : 74[FLAG]27[UID][BLOCK NO]

– 

, DUALi RW Protocol : 74[FLAG]28[UID]

– 

, DUALi RW Protocol : 74[FLAG]2C[UID][BLOCK NO][BLOCK COUNT]

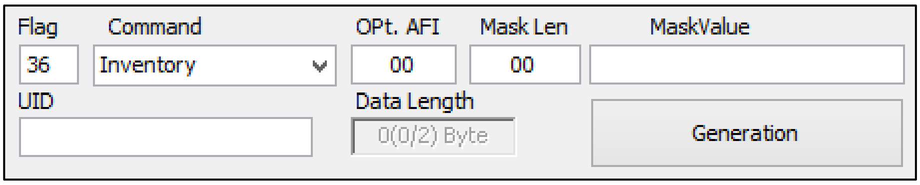
– 

, DUALi RW Protocol : 74[FLAG]29[UID][BLOCK NO]

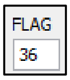
– 

, DUALi RW Protocol : 74[FLAG]2A[UID]

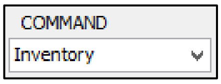
9.3 Command Execution



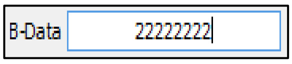
<Figure 9-4>

– 

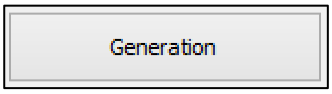
: Set flags for when the device sends commands to card

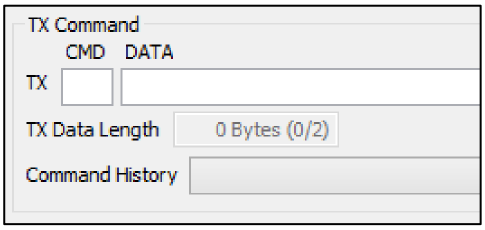
– 

: Select commands, this optional. Otherwise, the “Function Button” group box has a list ISO15693 commands, refer to <Figure9-3>.

– 

: The field where commands display responses.

- 

: Click after setting.
- 

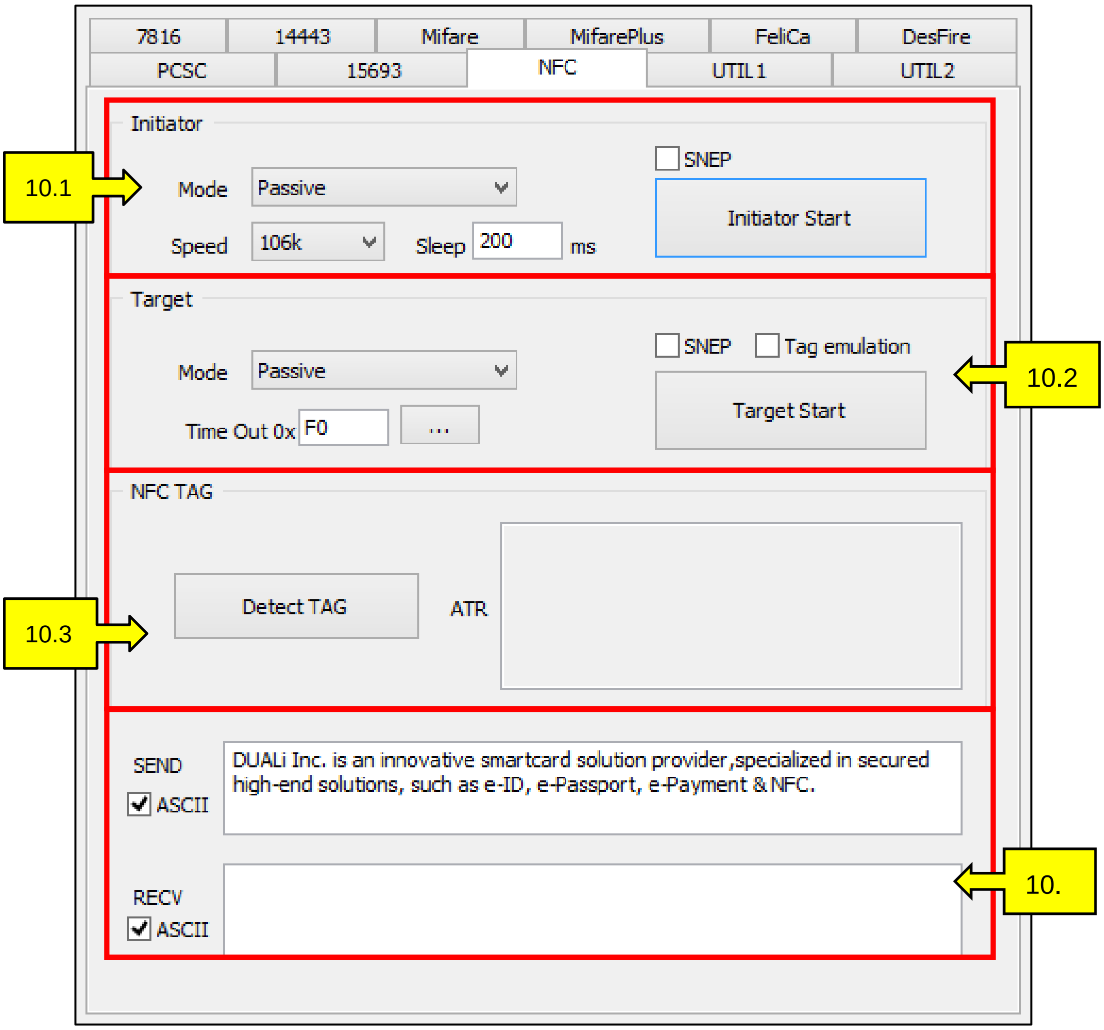
: the command frame will be displayed in the log, <Figure 1-7>.
- 

: Click to execute commands.

10. NFC

This tab, <Figure 10-1>, supports communication with ISO 18092 (Near Field Communication). ISO 18092 supports communication between Device to Device / Device to Card. Currently the communication speed supports 106, 212, and 424 kbps. For speeds of 106 kbps, the communication format is similar to that of ISO 14443 type A. For speeds of 212 and 424 kbps, the communication format is similar to that of FeliCa.

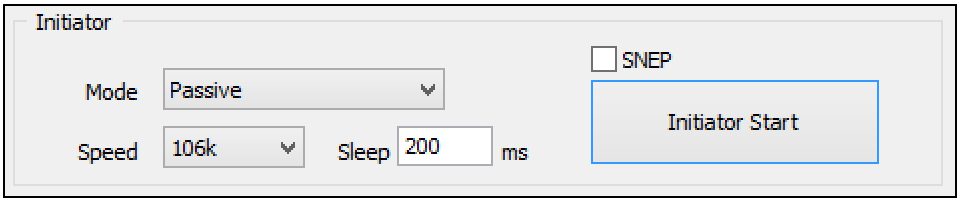
- Passive Communication: When one device creates an RF Field and the communication between devices are preset as either Initiator (creates RF field) or Target (receives power from Initiator).
- Active Communication: When both of the communication devices can create RF Fields and both devices can either be the Initiator or Target. The device which starts the communication is the Initiator and the other device is the Target.



<Figure 10-1>

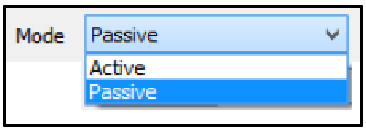
NFC Tab

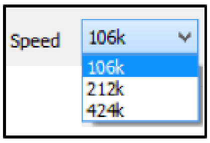
10.1 Initiator



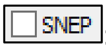
<Figure 10-2>

NFC Initiator

- An Initiator is the device that initiates the communication between NFC devices.
- 

: Select the communication mode of the Initiator.
- 

: Select the communication speed between the Initiator and Target.
- 

: Set the response waiting time.
- 

: Check the box to send commands in SNEP format.
- 

  DualCardDll API :

  LLC_Send_Text([PORT],0,[SLEN],[SBUF],[RLEN],[RBUF]);

  DE_NFC_Init_INITIATOR([PORT],[MODE],[SPEED],[RLEN],[RBUF]);

  DE_NFC_SendData([PORT],[NFC_INITIATOR_MODE],[SLEN],[SBUF],[RLEN],[RBUF]);

  DUALi RW Protocol :

  BC00[DATA]

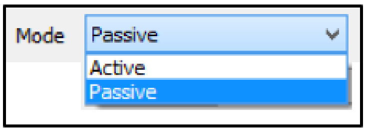
  94[MODE][SPEED]

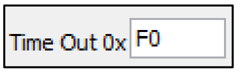
  96[DATA]

10.2 Target

<Figure 10-3>

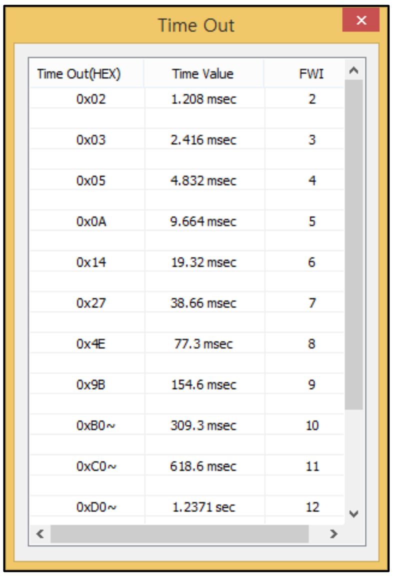
NFC Target

- A Target is the device that responds to the Initiator.
- 

: Select the communication mode of the Target.
- 

: Set the waiting time between sending commands and getting a response.
- 

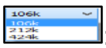
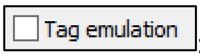
: Click to get more information. The Time Out box, <Figure 10-4>, will appear.



Time Out(HEX)	Time Value	FWI
0x02	1.208 msec	2
0x03	2.416 msec	3
0x05	4.832 msec	4
0x0A	9.664 msec	5
0x14	19.32 msec	6
0x27	38.66 msec	7
0x4E	77.3 msec	8
0x9B	154.6 msec	9
0xB0~	309.3 msec	10
0xC0~	618.6 msec	11
0xD0~	1.2371 sec	12

<Figure 10-4>

Time Out

- 
: Check the box to use the SNEP mode
- 

: Check the box to use the Tag Emulation mode

– 

DualCardDll API :

DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF],[TOUT]);

LLC_Send_Text([PORT],[MODE],[SLEN],[SBUF],[RLEN],[RBUF]);

DE_NFC_Init_TARGET([PORT],[TIMEOUT],[MODE],[RLEN],[RBUF]);

DE_NFC_GetTargetData([PORT],[RLEN],[RBUF]);

DE_NFC_SetTargetData([PORT],[SLEN],[SBUF]);

DE_NFC_GetTargetData([PORT],[RLEN],[RBUF]);

DUALi RW Protocol :

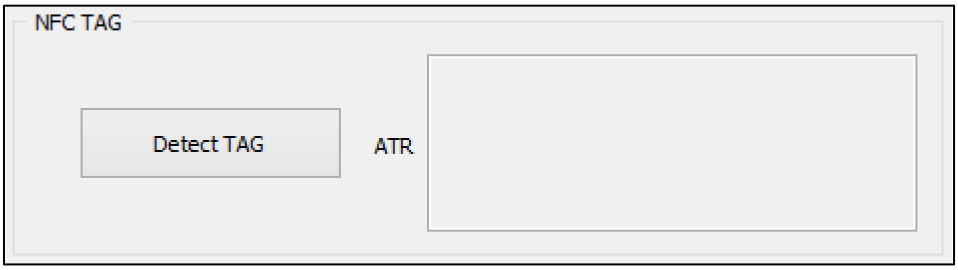
BB[TAG TYPE][DATA]

BC[MODE=20][DATA]

9574[MODE]00[TIMEOUT]

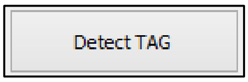
96[DATA]

10.3 NFC Tag



<Figure 10-5>

Detect NFC Tag

– 

: Click to detect tag type by ATR data 

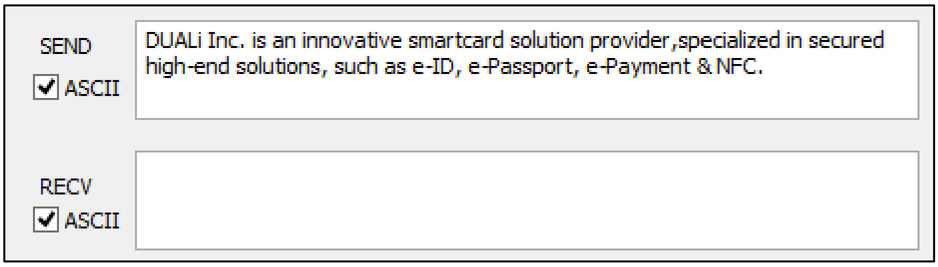
DualCardDll API :

DE_ByPassCommand(m_pDoc->m_nPort, nSLen, 0x83, SBUF, &outlen, RBUF);

DUALi RW Protocol :

83

10.4 Message Transmission



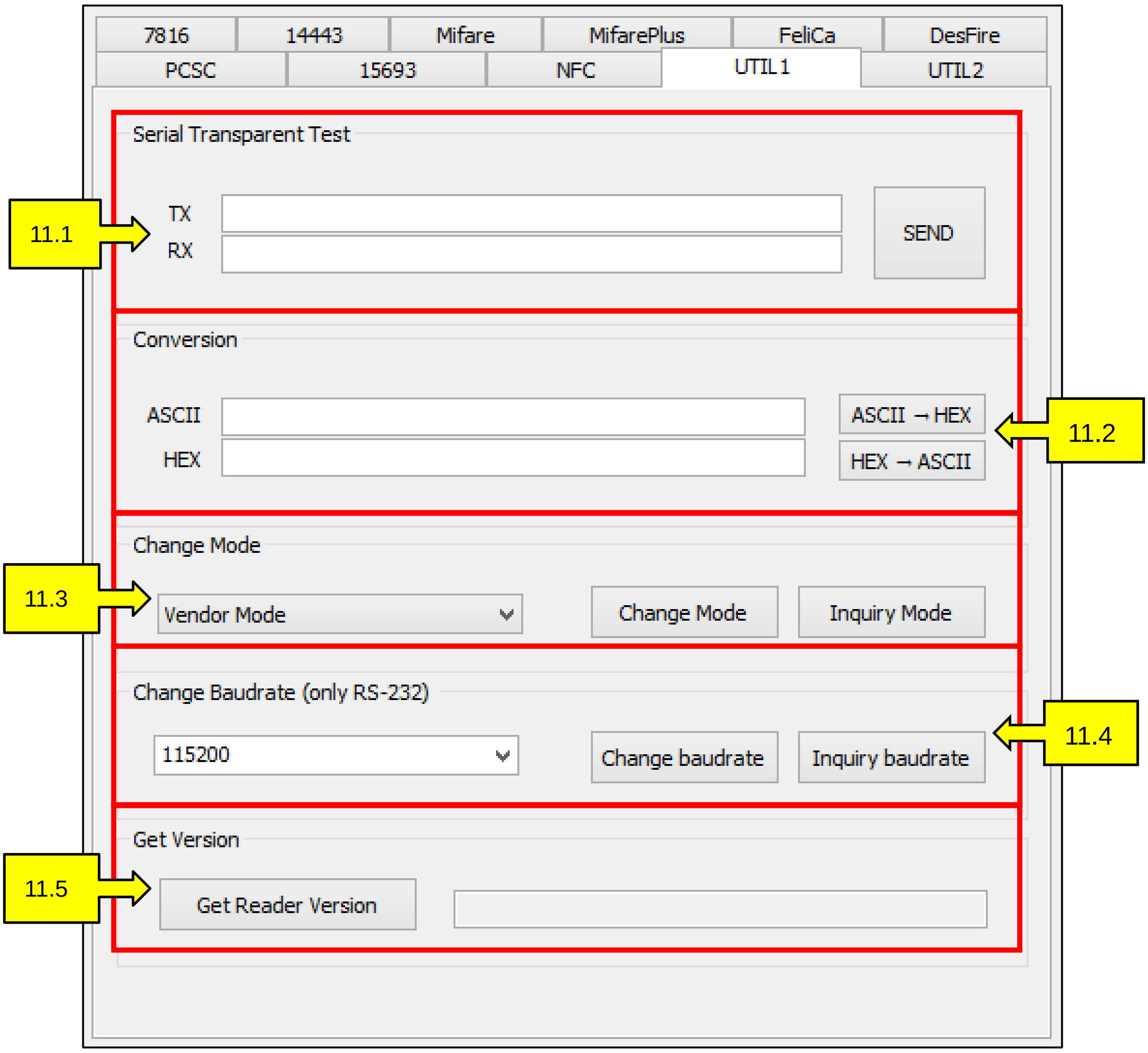
<Figure 10-6>

NFC Message Transmission

Use this group box to transmit messages, using ASCII or Hexa mode, to and from NFC devices.

11. UTIL1

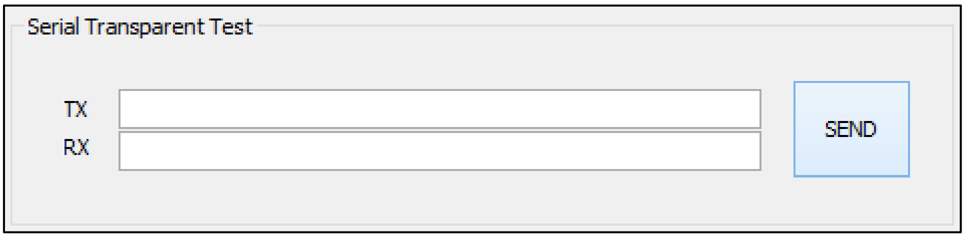
This tab, <Figure 11-1>, has various functions for the user's convenience.



<Figure 11-1>

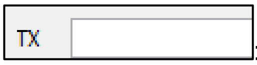
UTIL1 Tab

11.1 Serial Transparent Test

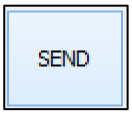


<Figure 11-2>

Serial Transparent Test

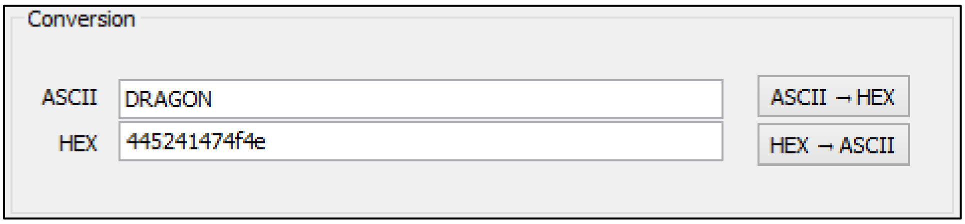
- This group box tests the sending and receiving of data via UART.
- 

: Input the command/data to send.
- 

: The response data from the serial equipment is displayed.
- 

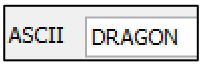
: Click to send input data.

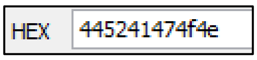
11.2 Conversion Data Type

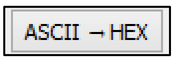


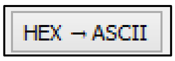
<Figure 11-3>

UTIL1 Conversion

- This function is used to convert from a ASCII value to a HEX value or from a HEX value to a ASCII value.
- 

: Input the ASCII value
- 

: Input the HEX value
- 

: Click to convert from a ASCII value to a HEX value
- 

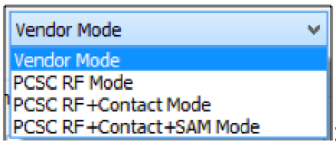
: Click to convert from a HEX value to a ASCII value.

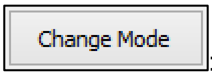
11.3 Change Mode

Users can easily change from Vendor Mode to PCSC Mode, or vice versa, when using USB connected devices.

<Figure 11-4>

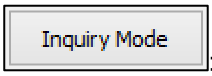
Change Modes

- 

: Select the mode to change to
- 

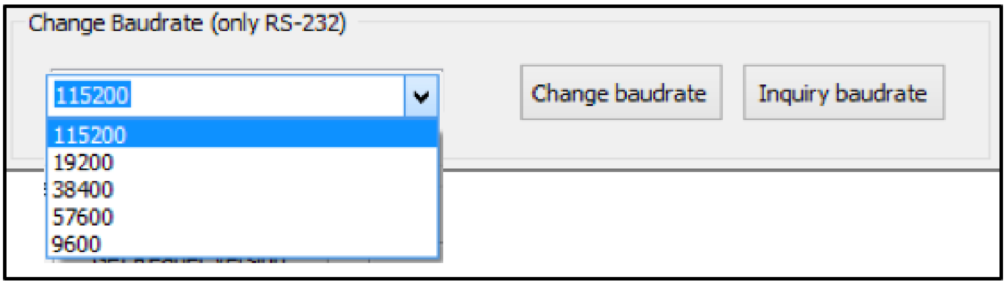
: Click to execute the change
- If the PCSC driver is not yet installed, you have to install the driver after you change to the PCSC mode.

  DualCardDll API : DE_ChangeDevice([SBUF],[MODE],[INQ FLAG]);

  DUALi RW Protocol : 1501[MODE]01
- 

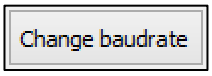
: Click to view the current mode of the connected device

11.4 Change Baudrate



<Figure 11-5>

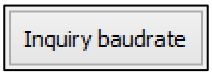
Change Baudrate

- 

:

DualCardDll API : DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF],[TOUT]);

DUALi RW Protocol : 1502[BAUD]

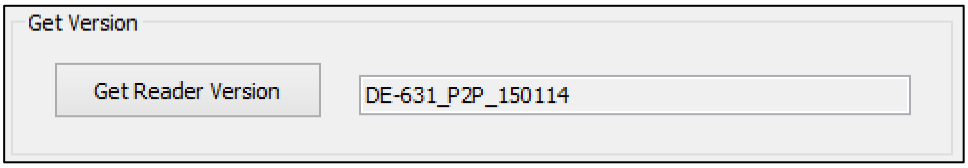
– 

:

DualCardDll API : DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF],[TOUT]);

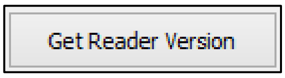
DUALi RW Protocol : 1503

11.5 Get Version



<Figure 11-6>

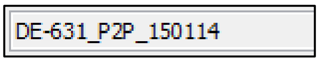
Reader Version

– 

: Click to get the reader’s version.

DualCardDll API : DE_GetVersion([PORT],[RLEN],[RBUF]);

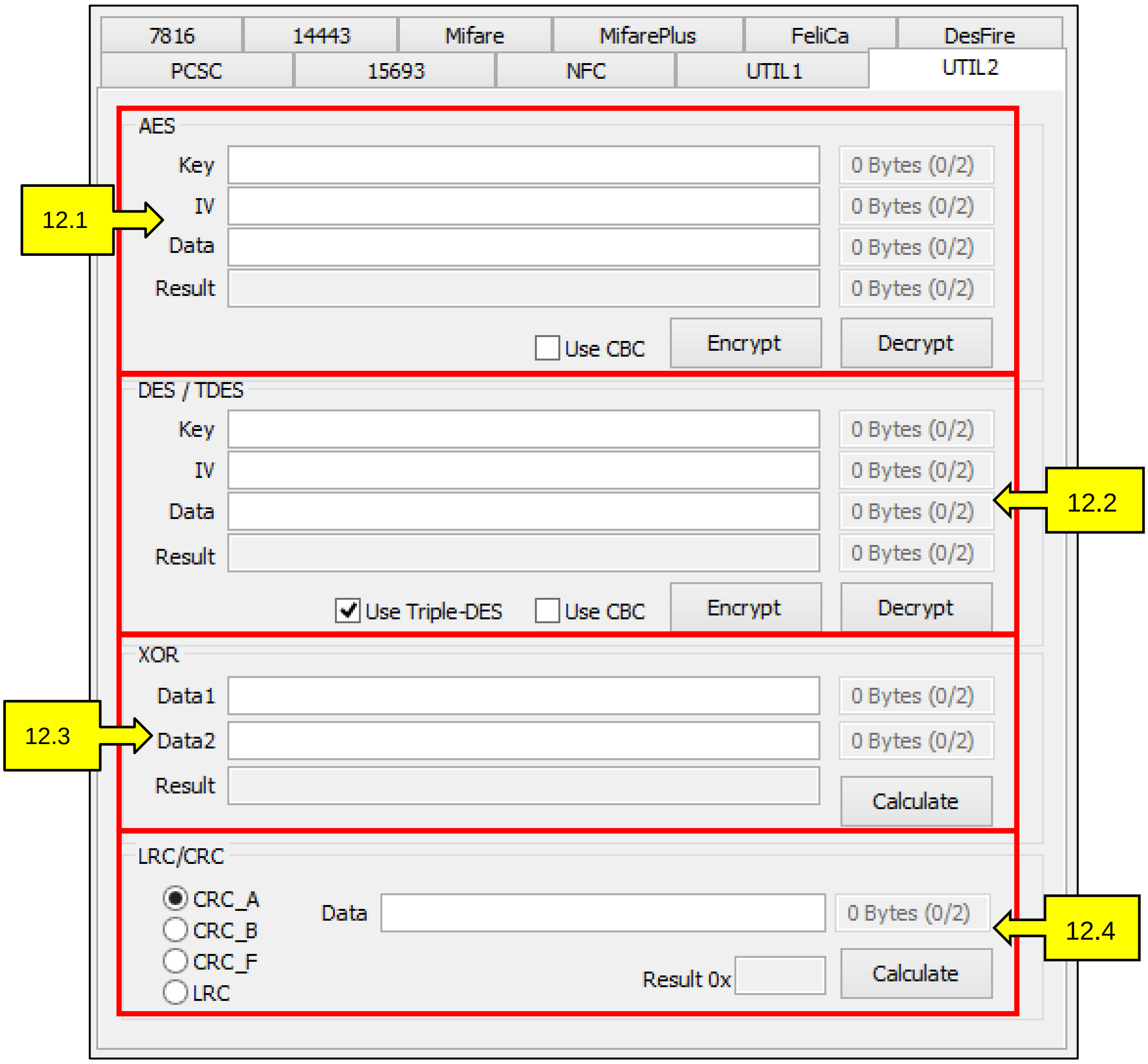
DUALi RW Protocol : 16

– 

: The reader’s version will be displayed in the field.

12. UTIL2

By using DualCard, users can check encrypted and decrypted values of AES, DES, and TDES algorithms. In this tab, there are XOR, LRC, and CRC calculation functions for the user's convenience.

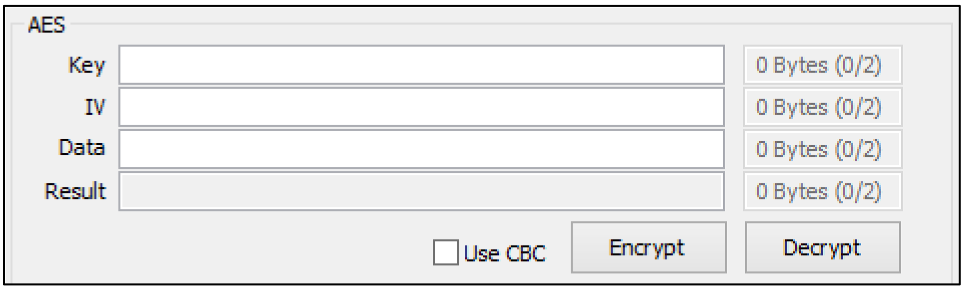


<Figure 12-1>

UTIL2 Tab

12.1 AES

This function is used to check encrypted and decrypted data results of an AES algorithm when using a certain key. Use 16 bytes units.



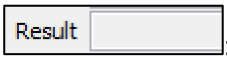
<Figure 12-2>

AES Algorithm

- 

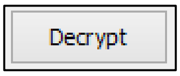
: Input Key.
- 

: Input Initial Vector in CBC mode.
- 

: Input data to encrypt and decrypt. To encrypt data, input plain data. To decrypt data, input encrypted data.
- 

: The encrypted and decrypted results will be displayed.
- 

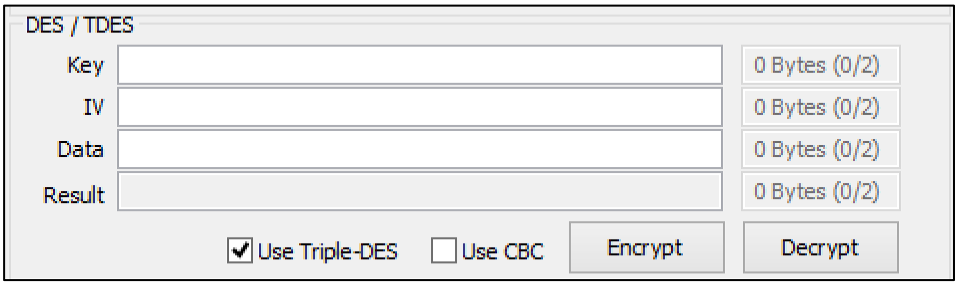
: Check the box to use the CBC mode. When unchecked, the mode is ECB.
- 

: Click to encrypt the input data.
- 

: Click to decrypt the input data.

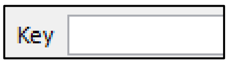
12.2 DES / TDES

This function is to check the results of data after the TDES algorithm. DES operates with 16 byte units and TDES operates with 8 byte units.



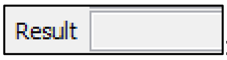
<Figure 12-3>

DES / TDES Algorithms

- 

: Input Key.
- 

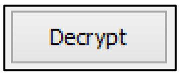
: Input initial vector in CBC mode.
- 

: Input data to encrypt and decrypt. To encrypt data, input plain data. To decrypt data, input encrypted data.
- 

: The encrypted and decrypted results will be displayed.
- 

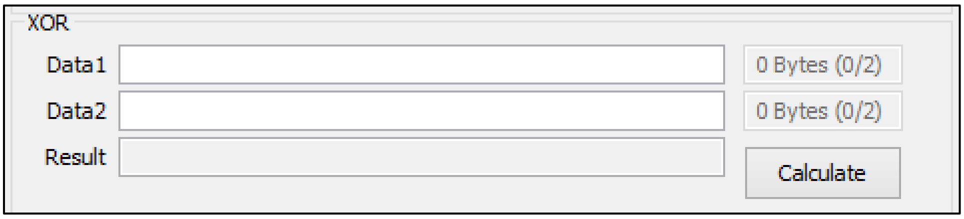
: Check the box to run TDES. If the box is not checked, the system will run DES.
- 

: Check the box to use the CBC mode. If the box is not checked the system will use the ECB mode.
- 

: Click to encrypt plain data. Results will populate.
- 

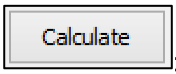
: Click to decrypt encrypted data. Results will populate.

12.3 Exclusive OR



<Figure 12-4>

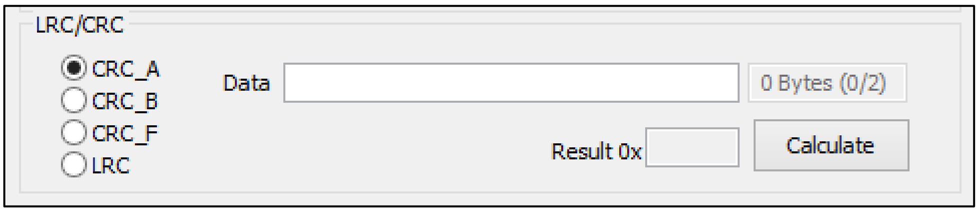
Exclusive OR

- To calculate the XOR of either Data1 or Data2, double click either of the two empty fields.
- 

: Click to calculate XOR. Results will display in the "Result" field.

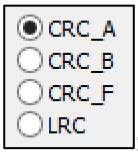
12.4 LRC / CRC

The LRC / CRC function is used to check for redundancy in send/receive data.



<Figure 12-5>

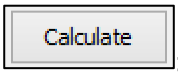
LRC/CRC

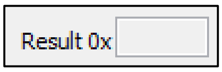
– 

: Select mode to check for redundancy.

– 

: Insert data

– 

: Click to show results in the field, 

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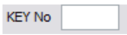
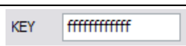
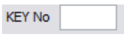
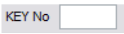
13. Annex

13.1 Mifare Commands

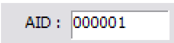
Command	Explanation
Auto Read	Read card automatically DualCardDll API : DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF] ,[TOUT]) DUALi RW Protocol : 3600[AUTH MODE][BLOCK NO][KEY]; 27[BLOCK NO]
Auto write	Write card automatically DualCardDll API : DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF] ,[TOUT]) DUALi RW Protocol : 3800[AUTH MODE][BLOCK NO][KEY][DATA] 28[BLOCK NO][DATA]
Decrement Transfer	Decrement the value and transfer to selected block DualCardDll API : DEA_Dec_Transfer([PORT],[BLOCK NO],[VALUE],[TRANSFER BLOCK NO]); DUALi RW Protocol : 2C[BLOCK NO][VALUE][TRANSFER BLOCK NO]
Decrement Transfer, Verify by Reader	Decrement to transfer and verify by reader DualCardDll API : DEA_Dec_Transfer2([PORT],[BLOCK NO],[VALUE],[TRANSFER BLOCK NO]); DUALi RW Protocol : 34[BLOCK NO][VALUE][TRANSFER BLOCK NO]
Increment Transfer	Increment to transfer. DualCardDll API : DEA_Inc_Transfer([PORT],[BLOCK NO],[VALUE],[TRANSFER BLOCK NO]); DUALi RW Protocol : 2B[BLOCK NO][VALUE][TRANSFER BLOCK NO]
Increment Transfer, Verify by Reader	Increment to transfer and verify by reader. DualCardDll API : DEA_Inc_Transfer2([PORT],[BLOCK NO],[VALUE],[TRANSFER BLOCK NO]); DUALi RW Protocol : 33[BLOCK NO][VALUE][TRANSFER BLOCK NO]
Loadkey	Load key data to reader DualCardDll API : DEA_Loadkey([PORT],[MODE],[KEY NO],[KEY]); DUALi RW Protocol : 2F[MODE][KEY NO][KEY]
Request Auth	Request, Anticollision, Select, Authentication with  DualCardDll API : DEA_Req_Auth([PORT],[REQUEST MODE],[AUTH MODE],[KEY NO], [BLOCK NO],[RLEN],[RBUF]); DUALi RW Protocol : 3100[AUTH MODE][KEY NO][BLOCK NO]
Request Authkey	Request, Anticollision, Select, Authentication with  DualCardDll API : DEA_Req_Authkey([PORT],[REQUEST MODE],[AUTH MODE],

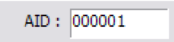
	[BLOCK NO],[KEY],[RLEN],[RBUF]); DUALi RW Protocol : 3200[AUTH MODE][BLOCK NO][KEY]
Request to Read with Key No	Request to read with  DualCardDll API : DEA_Req_AuthRead([PORT],[REQUEST MODE],[AUTH MODE],[KEY NO],[BLOCK NO],[RLEN],[RBUF]); DUALi RW Protocol : 3500[AUTH MODE][KEY NO][BLOCK NO]
Request to Read with KeyData	Request to read with  DualCardDll API : DEA_Req_AuthkeyRead([PORT],[REQUEST MODE],[AUTH MODE], [BLOCK NO],[KEY],[RLEN],[RBUF]); DUALi RW Protocol : 3600[AUTH MODE][BLOCK NO][KEY]
Request to Select	Request to select DualCardDll API : DEA_Req_Select([PORT],[REQUEST MODE],[RLEN],[RBUF]); DUALi RW Protocol : 3900
Request to Write with Key No	Request to write with  DualCardDll API : DEA_Req_AuthWrite([PORT],[REQUEST MODE],[AUTH MODE],[KEY NO],[BLOCK NO],[DATA],[RLEN],[RBUF]); DUALi RW Protocol : 3700[AUTH MODE][KEY NO][BLOCK NO][DATA]
Request to Write with KeyData	Request to write with  DualCardDll API : DEA_Req_AuthkeyWrite([PORT],[REQUEST MODE],[AUTH MODE],[BLOCK NO],[KEY],[DATA],[RLEN],[RBUF]); DUALi RW Protocol : 3800[AUTH MODE][BLOCK NO][KEY][DATA]
Write for Mifare Ultra Light	Write command for Mifare Ultra Light DualCardDll API : DEA_UltraM_Write([PORT],[ADDRESS],[DATA]); DUALi RW Protocol : 3B[ADDRESS][DATA]
Req Idle	Request idle command DualCardDll API : DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF] ,[TOUT]) DUALi RW Protocol : 21
Write and Read, Verify by App	Write and read Mifare and verify by application DualCardDll API : SecReqIdle OnBnClickedButtonAnticollision OnBnClickedButtonSelect SecAuthKey SecWrite SecRead DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF] ,[TOUT]) DUALi RW Protocol :
Write and Read, Verify by Reader	Write and read Mifare and verify by reader DualCardDll API : DE_Polling([PORT],[SLEN],[SBUF],[RLEN],[RBUF] ,[TOUT]) DUALi RW Protocol : AA[FROM BLOCK NO][TO BLOCK NO][KEY][DATA]

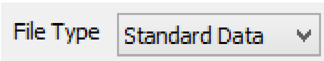
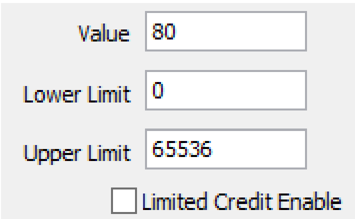
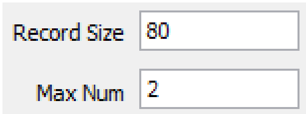
13.2 FeliCa Commands

Command	Explanation
Polling	Get a value of IDm and PMm
Polling-No Auth with RC-251	Get a value of IDm and PMm without RC-251 Auth
Mutual	Authentication command for traditional method
Read Block with ENC	Read command for encrypted Service
Write Block with ENC	Write command for encrypted Service
Read Block without ENC	Read command for a non-encrypted Service
Write Block without ENC	Write command for a non-encrypted Service
Read Block without ENC(Token)	Read command for a non-encrypted Service(Token)
Write Block without ENC(Token)	Write command for a non-encrypted Service(Token)
Increment with ENC	Increment command for encrypted Service
Decrement with ENC	Decrement command for encrypted Service
Cashback without ENC	Cashback command with non-encrypted Service
Read Balance	Read Balance

13.3 DesFire Commands

Command	Explanation
Create Application	This command allows the user to create new applications on the PICC. You need to set  to Create Application. DualCardDll API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN], [SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A020006CA[AID][KEY SET][KEY NO]
Delete Application	This command allows the user to permanently deactivate applications on the PICC. DualCardDll API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN], [SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A020004DA[AID]
Get Application IDs	This command returns the Application IDs of all active applications on a PICC. DualCardDll API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN], [SBUF],[RLEN],[RBUF]);

	DUALi RW Protocol : 3A0200016A
Get Key Setting	This command allows the user to get information on the PICC and application master key settings. In addition, it returns the maximum number of keys that can be stored in the selected application. DualCardDll API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN], [SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A02000145
Change Key Settings	This command changes the master key settings depending on the selected AID.  DualCardDll API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN], [SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A02000154[KEY SET]
Change Key	This command allows the user to change any key stored in the PICC. You need to input Key No in . DualCardDll API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN], [SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A02[FLAG]02C4[KEY NO][KEY]
Get Version	DualCardDll API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN], [SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A02000160
Get Available Free Memory	DualCardDll API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN], [SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A0200016E
Format Card	This command releases the PICC user memory. DualCardDll API :

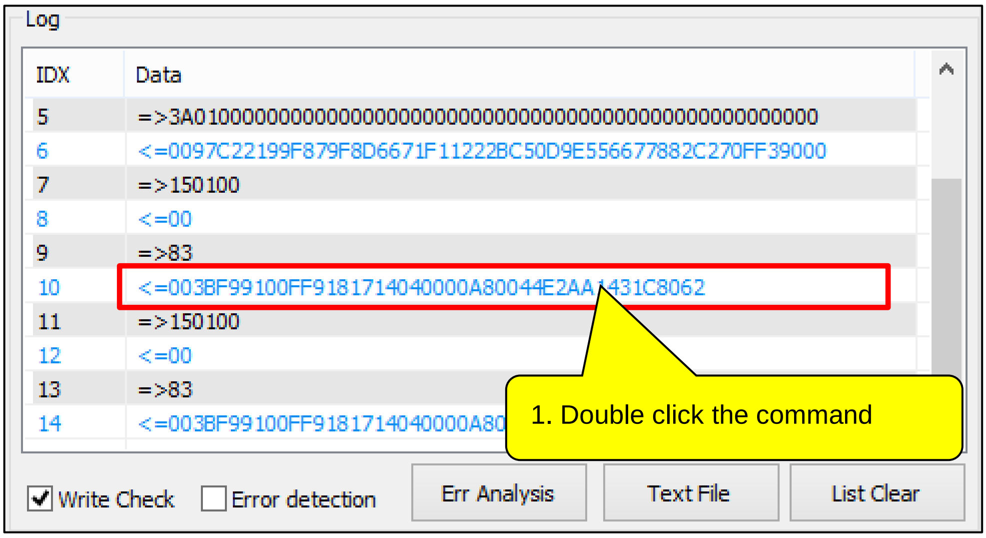
	DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN], [SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A020001FC
Commit Transaction	This command allows the user to validate all previous write access on backup data files, value files, and record files in one application. DualCardDll API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN], [SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A020001C7
Create File	This command creates files according to the selected file type  . (Standard Data, Backup Data, Value, Linear Record, Cyclic Record) You need to set one of options  ,  or  DualCardDll API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN], [SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A020008CD[FILE NO][COMM SET][ACCESS RIGHT][FILE SIZE] 3A020008CB[FILE NO][COMM SET][ACCESS RIGHT][FILE SIZE] 3A020008CC[FILE NO][COMM SET][ACCESS RIGHT][LOWER LIMIT] [UPPER LIMIT][VALUE] 3A020011C0[FILE NO][COMM SET][ACCESS RIGHT][RECORD SIZE][NUM RECORD] 3A020011C1[FILE NO][COMM SET][ACCESS RIGHT][RECORD SIZE][NUM RECORD]
Delete File	This command permanently deactivates a file within the file directory of the selected application. DualCardDll API :

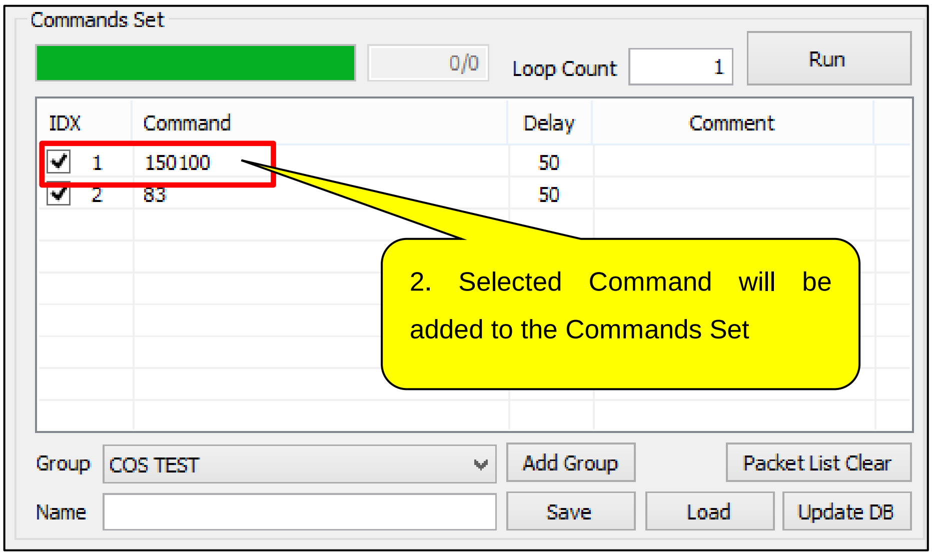
	DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN], [SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A020002DF[FILE NO]
Get File IDs	This command returns the File IDs of all active files in the selected application. DualCardDll API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN], [SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A0200016F
Get File Setting	This command allows the user to get information on the properties of a specific file. The information provided by this command depends on the type of file that is queried. DualCardDll API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN], [SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A0200025F[FILE NO]
Change File Settings	This command changes the access parameters of an existing file. DualCardDll API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN], [SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A020005F5[FILE NO][COMM SET][ACCESS RIGHT]
Change File Settings-Free Access	This command changes the access parameters of an existing file for free access. DualCardDll API : DE_DESFireTransparent([PORT],[FLAG],[CMDLEN],[SLEN], [SBUF],[RLEN],[RBUF]); DUALi RW Protocol : 3A020005F5[FILE NO][COMM SET]EEEE

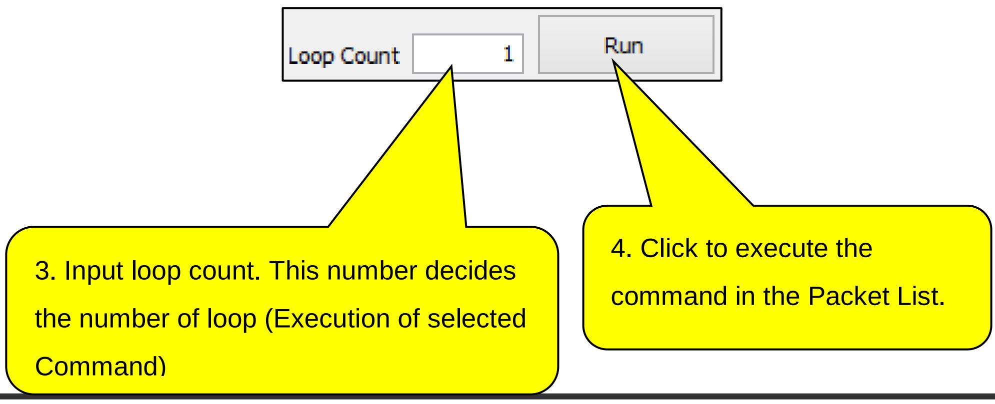
14. Example

14.1 Add Commands to the Packet List and Save/Load Command Sets

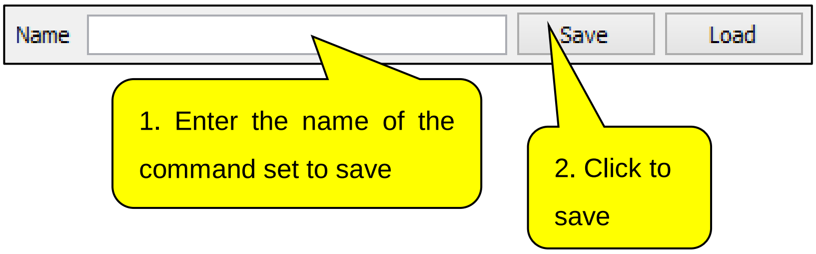
– Add Commands to the Packet List:



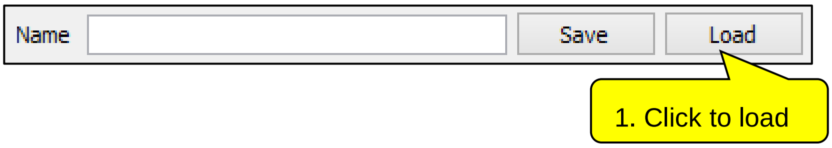


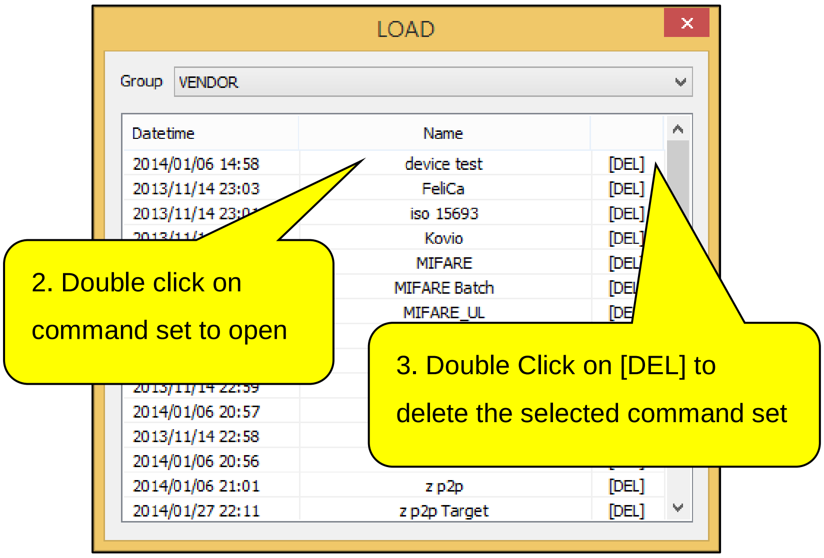


- Save Command Set



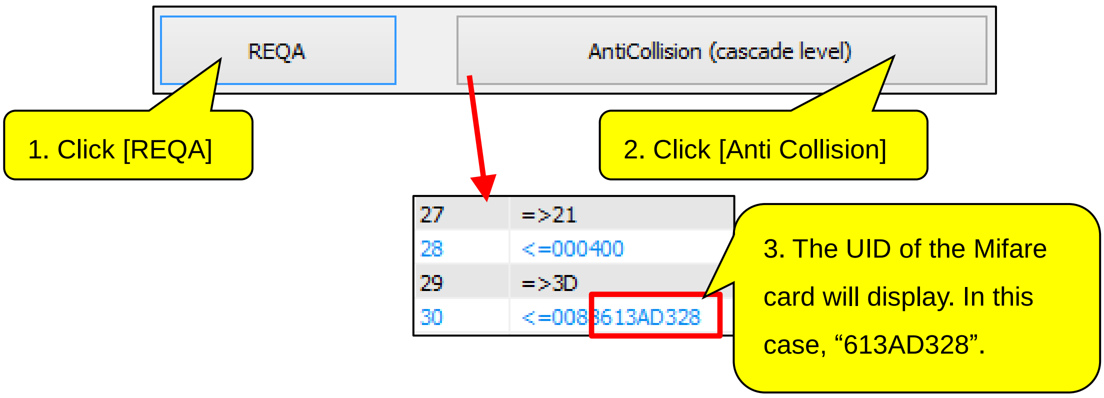
- Load/Delete Command Set

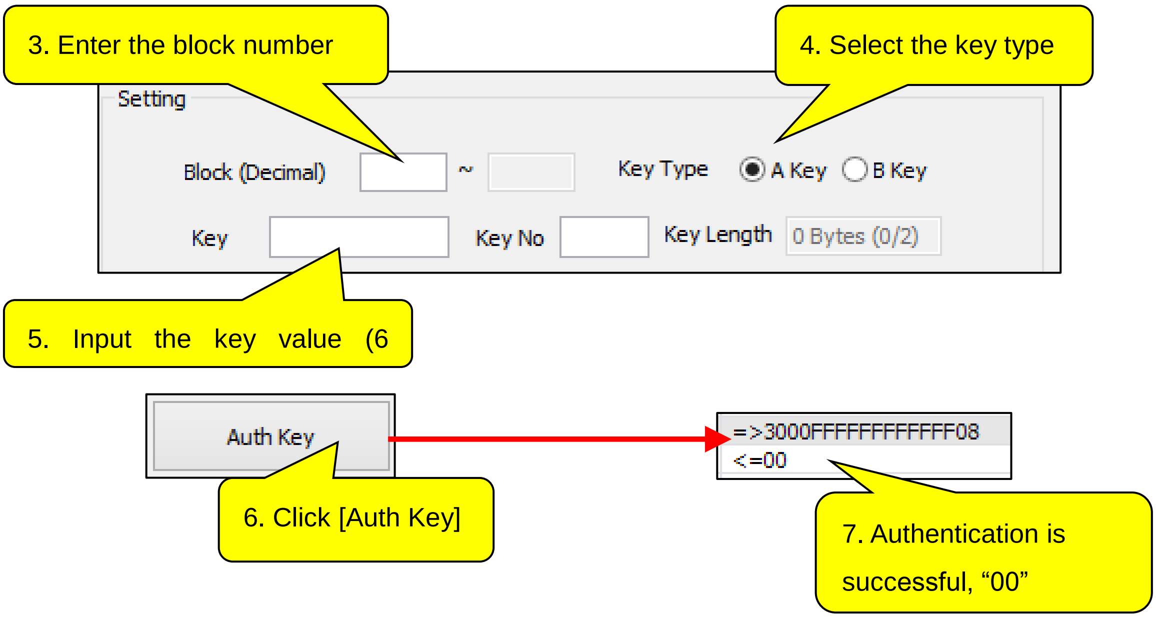




14.2 14.2 Mifare

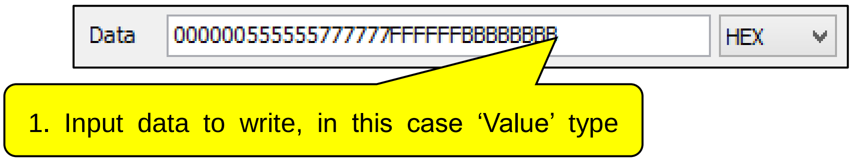
- [Auth Key] Authentication

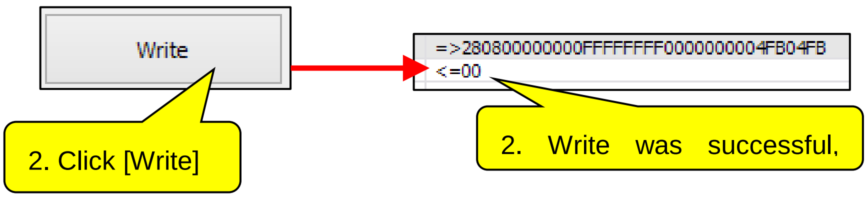




– [Write]

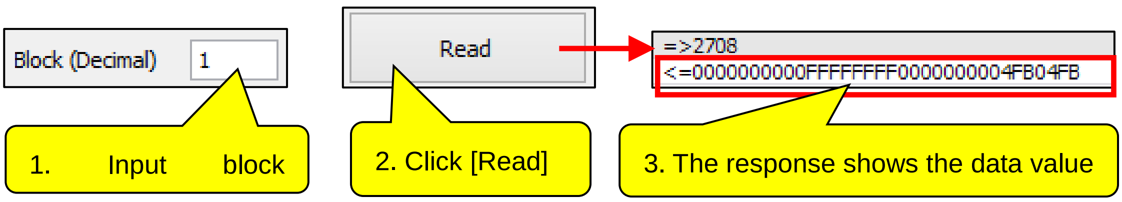
The write process has to be done after the [Auth Key] procedure





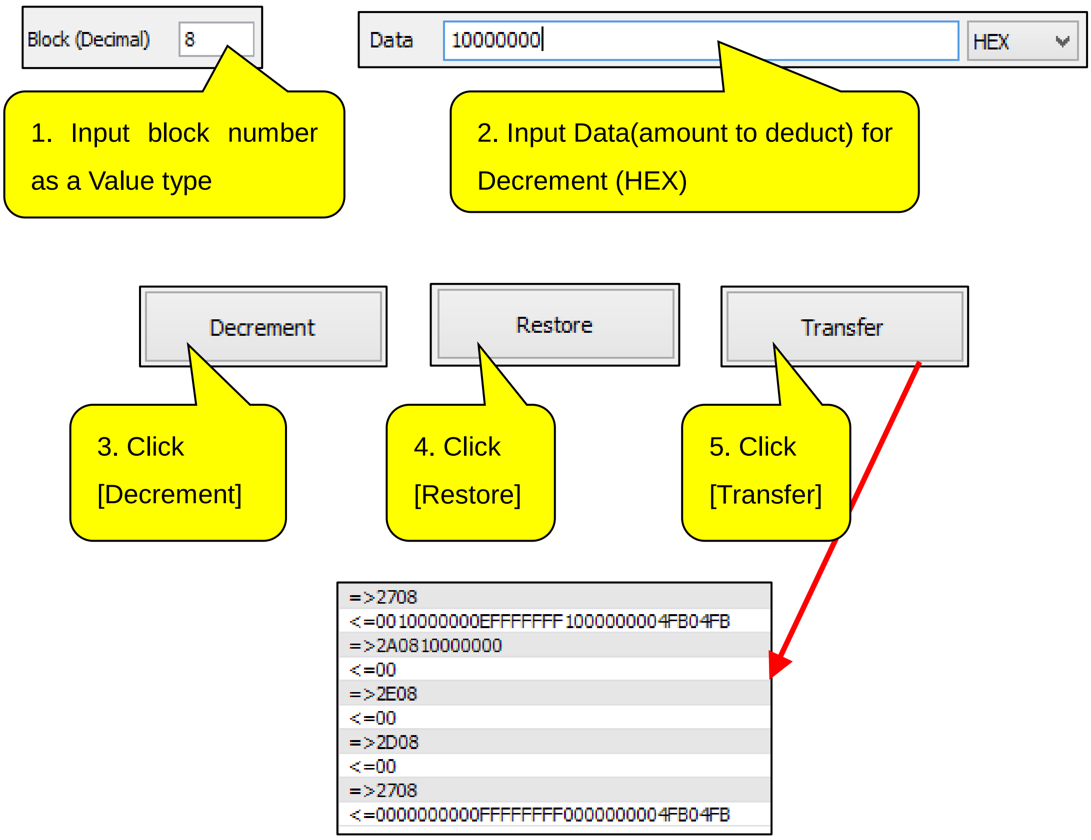
– [Read]

The read process has to be done after the [Auth Key] procedure



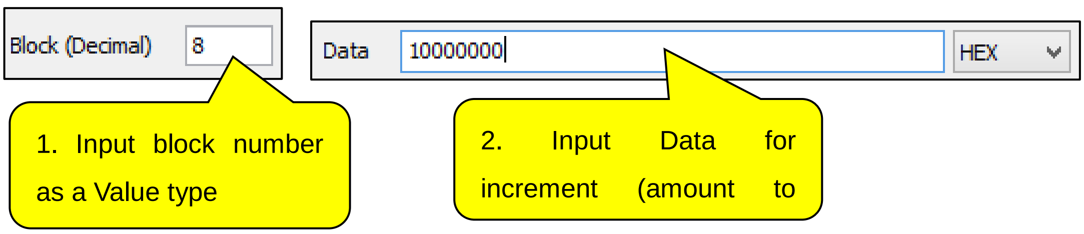
- [Decrement]

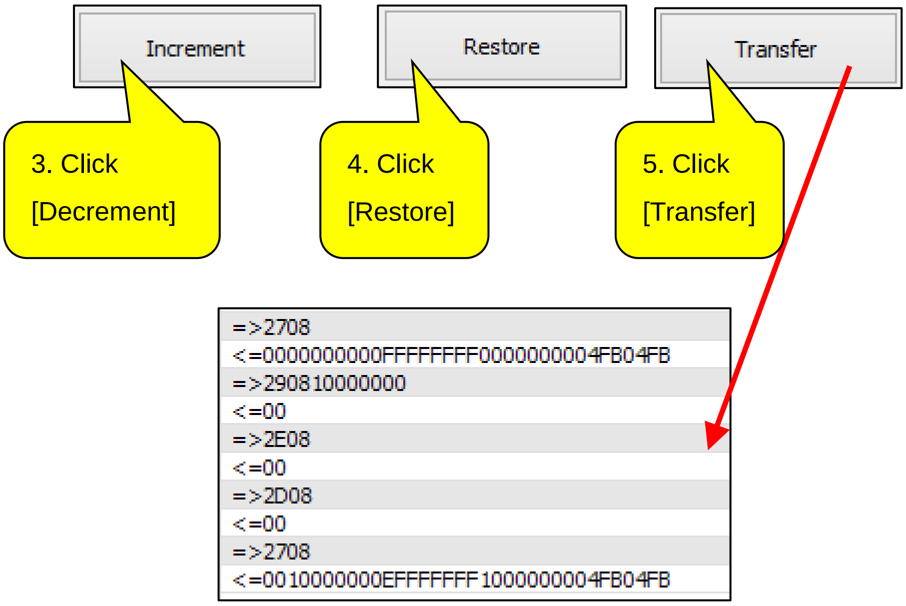
  The decrement process has to be executed after the [Auth Key] process. For decrement, the block data must be a Value type.



- [Increment]

  The Increment process has to be done after the [Auth Key] procedure. For Increment, block data must be a 'Value type'.

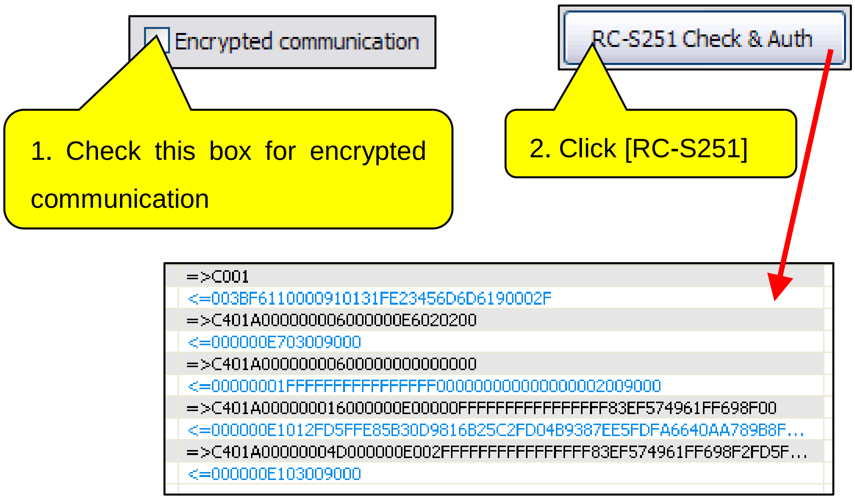


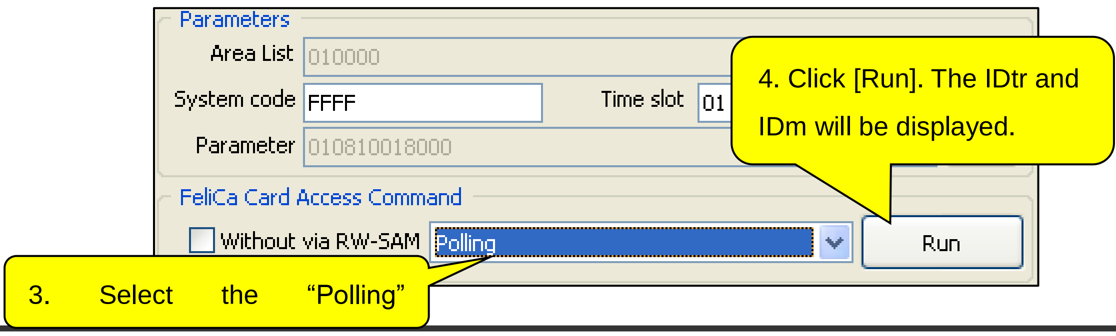


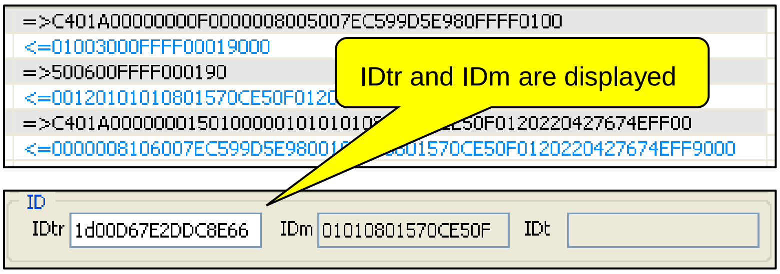
14.3 FeliCa

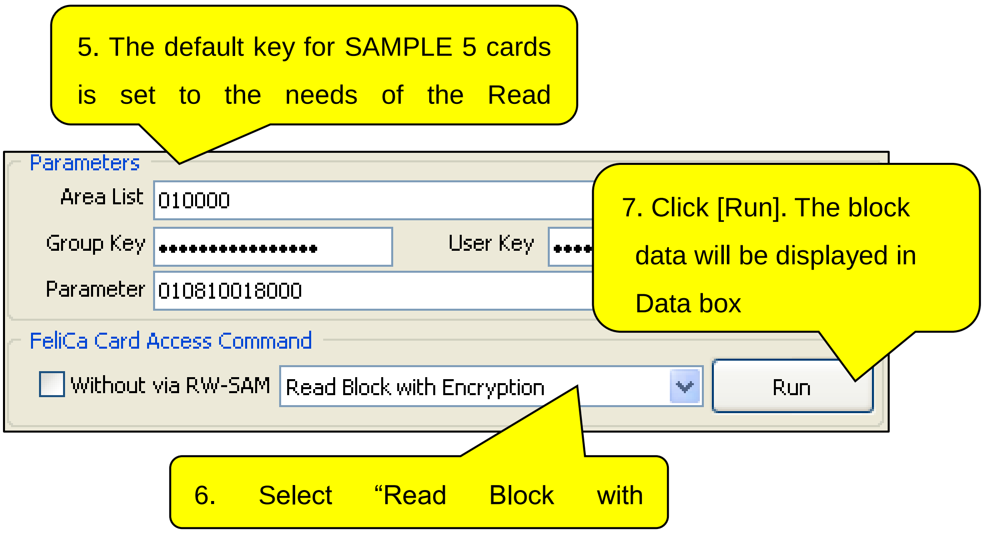
This procedure is based on FeliCa's SAMPLE 5 cards.

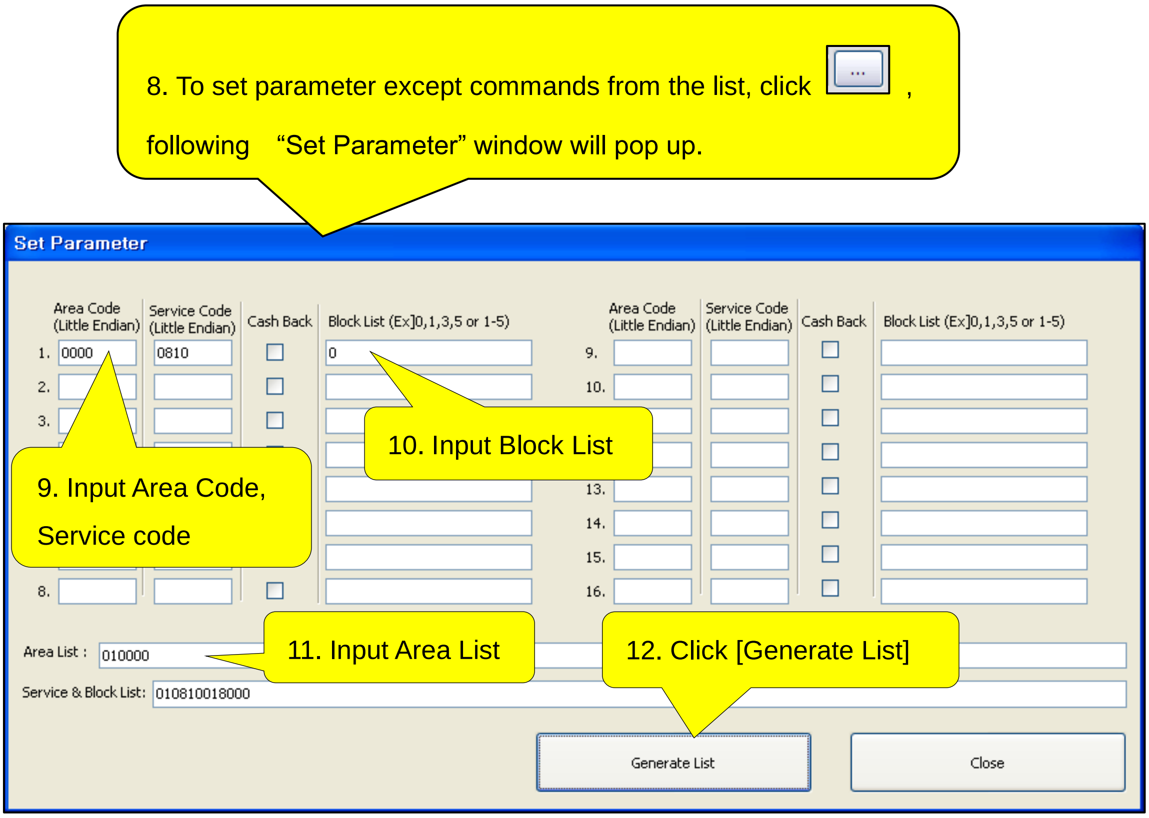
- [Read]

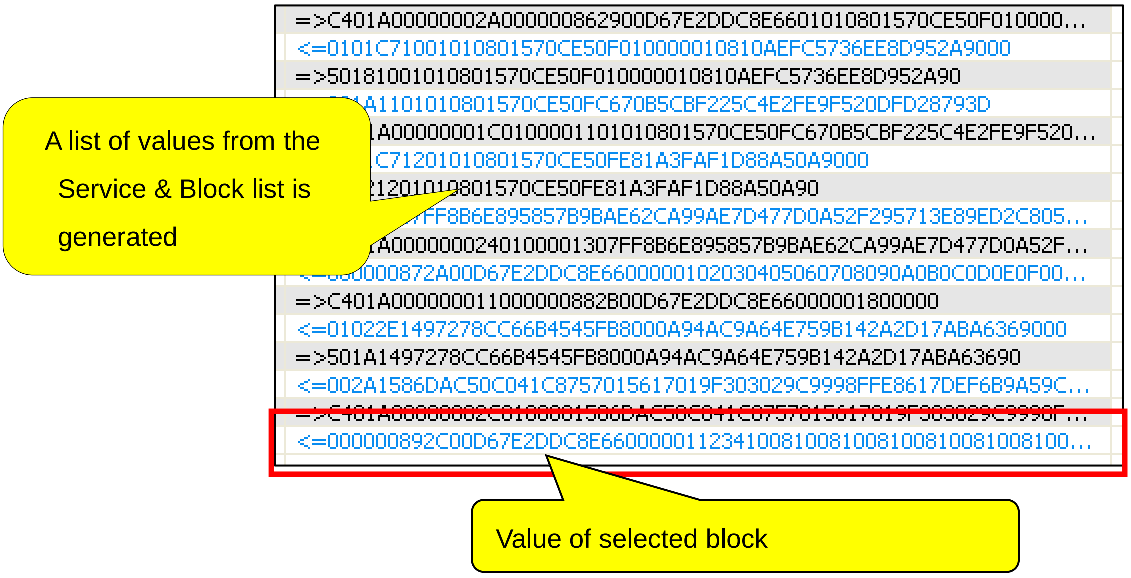





=>C401A00000000F0000008005007EC599D5E980FFFF0100
<=01003000FFFF00019000
=>500600FFFF000190
<=001201010108015700CE50F0120
=>C401A0000000150100000101010101
<=0000008106007EC599D5E98001
2220427674EFF9000
IDtr and IDm are displayed
ID
IDtr 1d00D67E2DDC8E66
IDm 01010801570CE50F
IDt

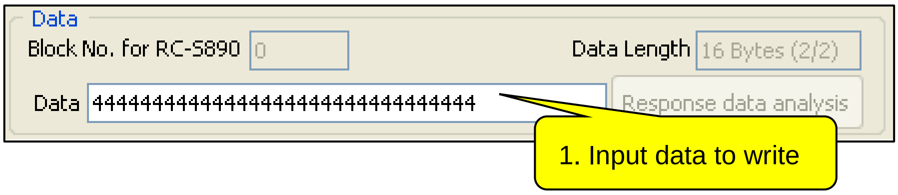

5. The default key for SAMPLE 5 cards is set to the needs of the Read
Parameters
Area List 010000
Group Key ••••••••••••••••
User Key ••••
Parameter 010810018000
7. Click [Run]. The block data will be displayed in Data box
FeliCa Card Access Command
Without via RW-SAM
Read Block with Encryption
Run
6. Select “Read Block with

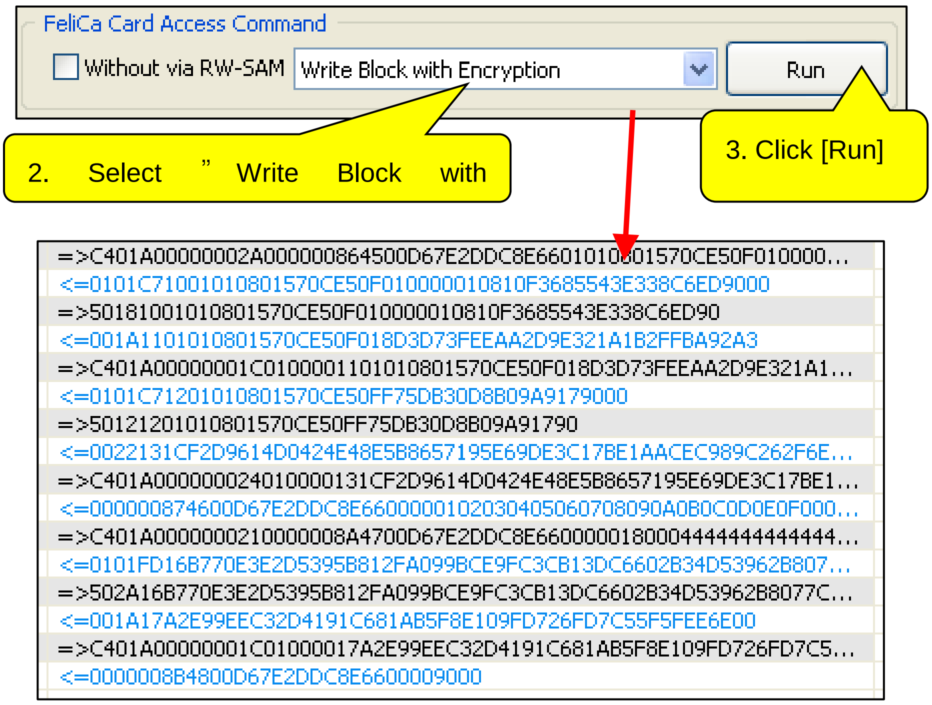

8. To set parameter except commands from the list, click ... , following “Set Parameter” window will pop up.
Set Parameter
Area Code (Little Endian)
Service Code (Little Endian)
Cash Back
Block List (Ex]0,1,3,5 or 1-5)
0000
0810
0
9. Input Area Code, Service code
10. Input Block List
11. Input Area List
12. Click [Generate List]
Area List : 010000
Service & Block List: 010810018000
Generate List
Close



- [Write]

The RW-SAM authentication process is required before the Write process.



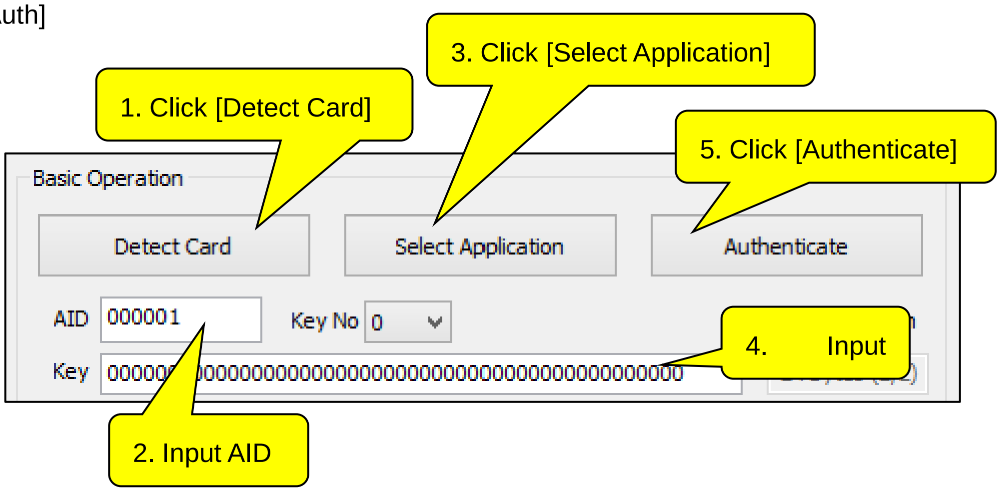


14.4 DesFire

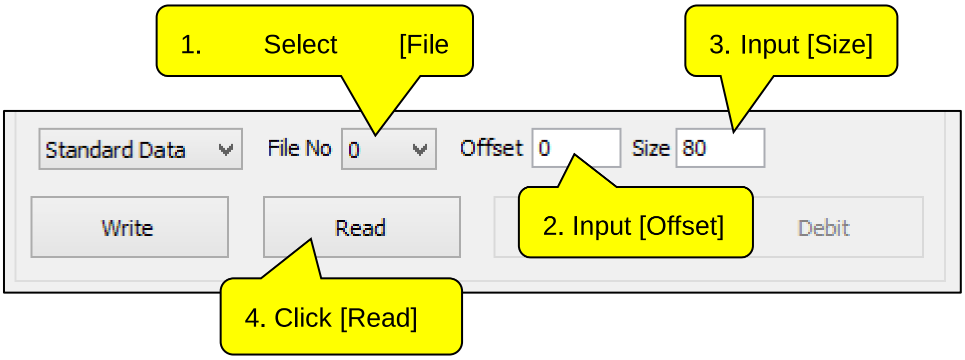
14.4.1 Basic Operation

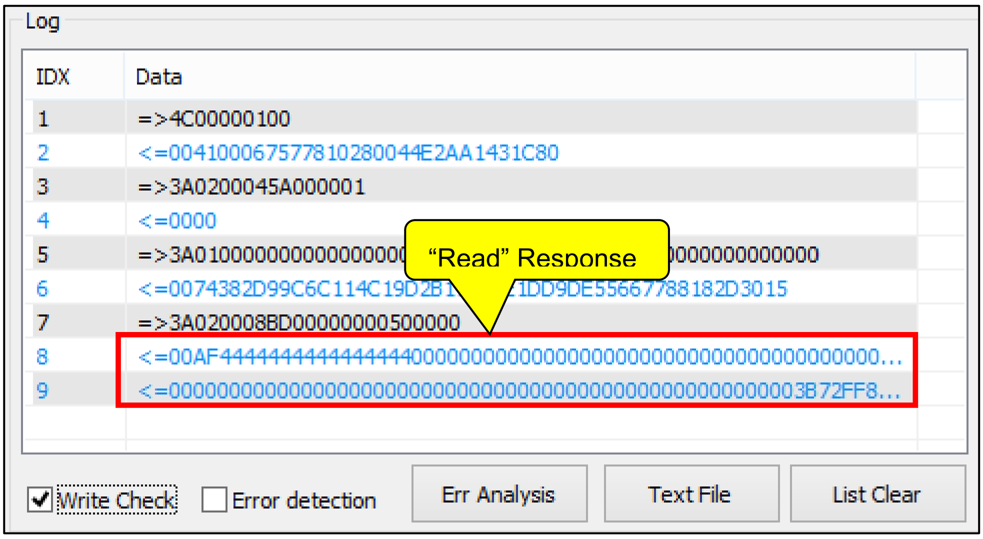
This procedure is based on the default key value and standard data file type. AID is '000001'

- [Auth]

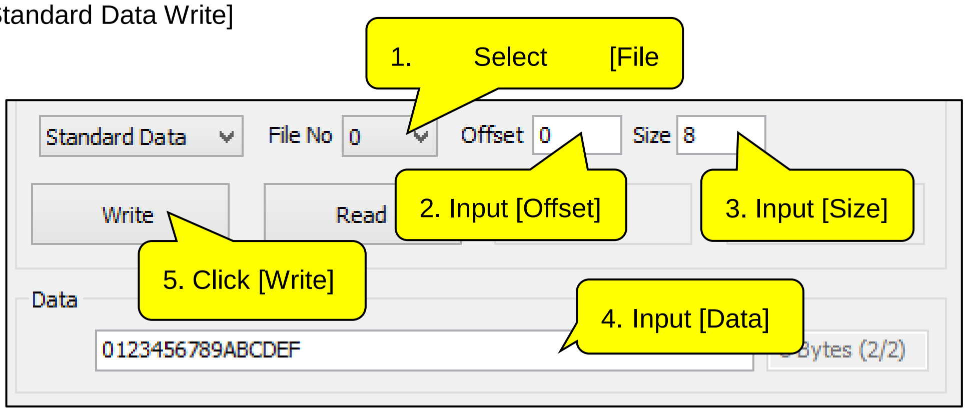


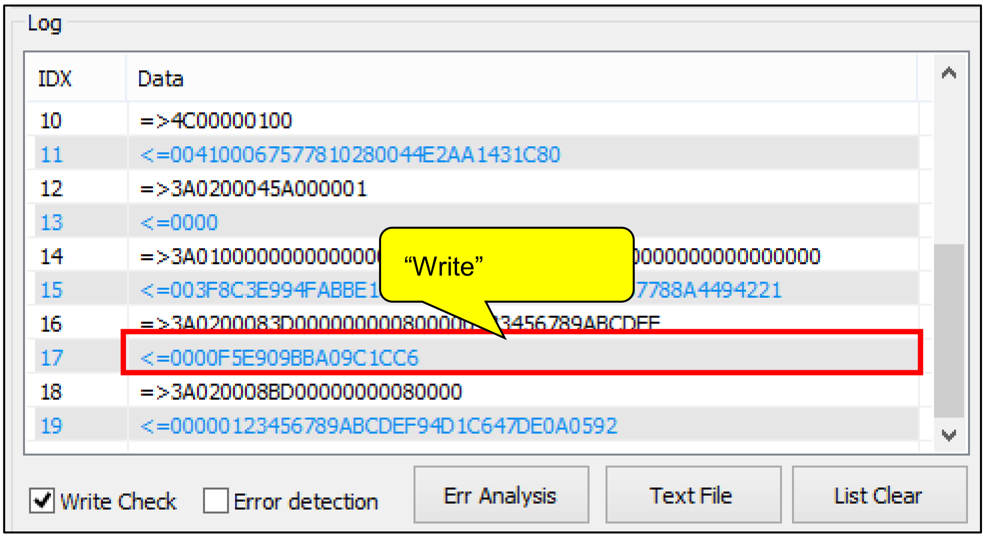
- [Standard Data Read]





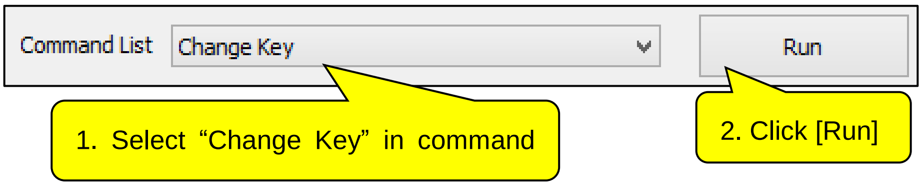
– [Standard Data Write]

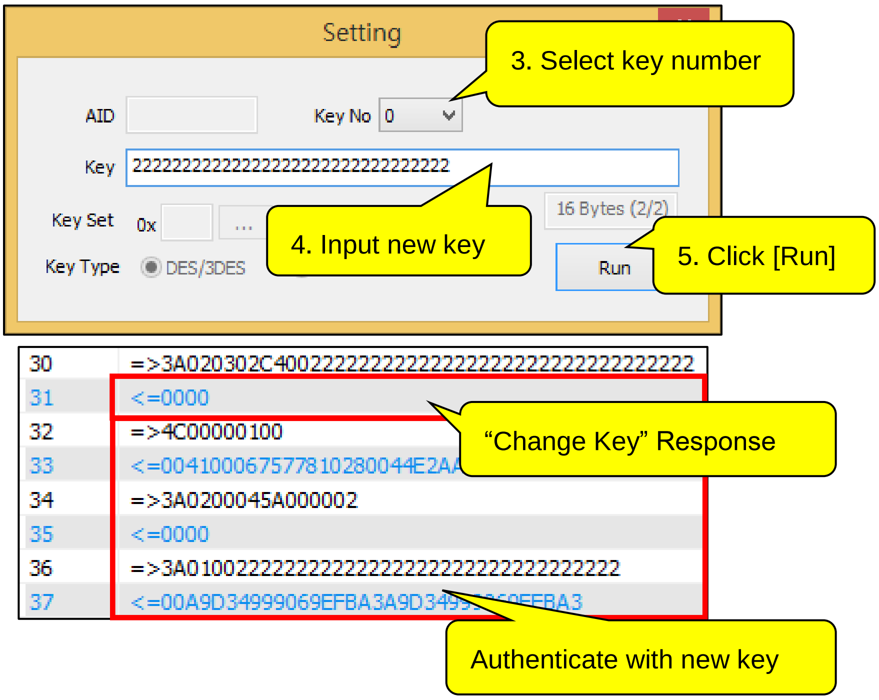




– [Change Master Key]

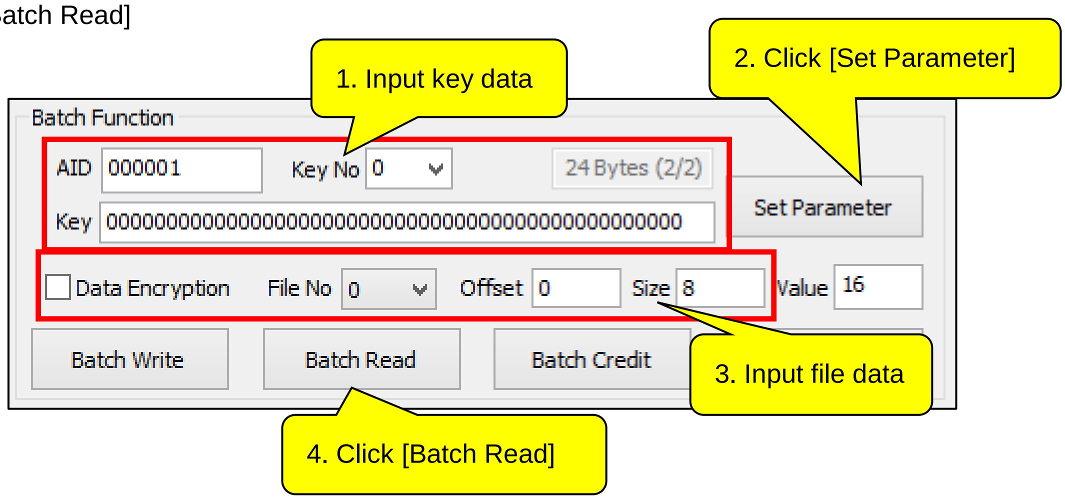
The Key '0' Authentication process is required before the Change Master Key process.

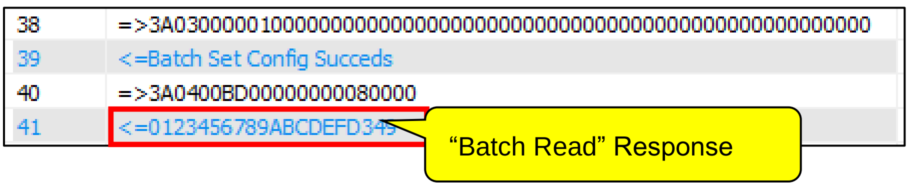




14.4.2 Batch Command

– [Batch Read]





− [Batch Credit]